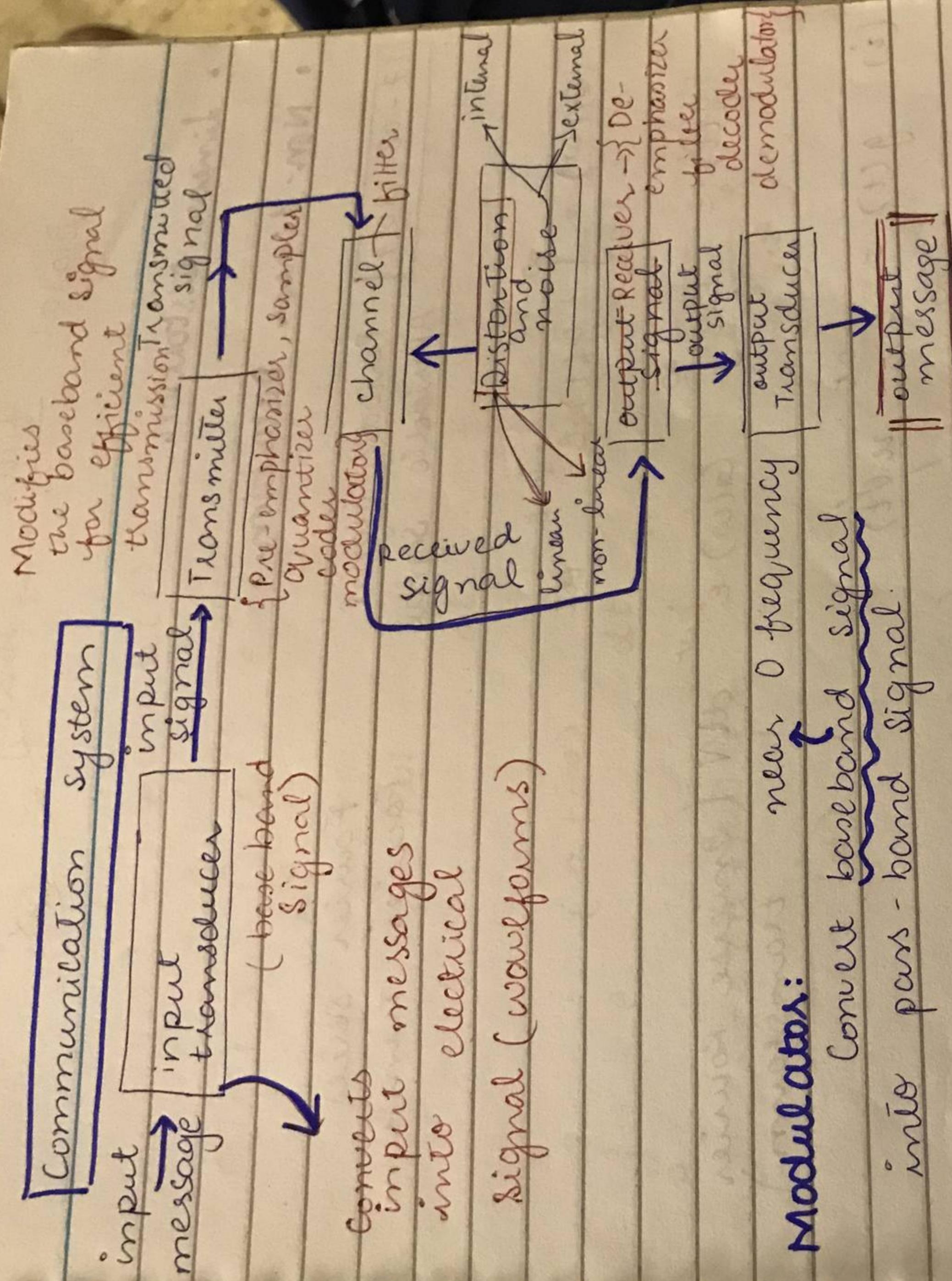


(3)



communication

$x(t) \rightarrow h(t) \rightarrow y(t)$

$$y(t) = x(t) * h(t)$$

$$Y(\omega) = X(\omega) \cdot H(\omega)$$

$$|Y(\omega)| = e^{j\theta Y(\omega)}$$

Digital ^{low pass filter} → Analog

(4)

- Linear Distortion

- Non-Linear Distortion

17-08-2017

Periodic Signal
Aperiodic Signal

Fourier Series
Fourier Transform.

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} d\omega \quad (\text{Inverse Fourier transform})$$

(i) $g(t) = e^{-at} u(t)$

$$G(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \frac{-1}{a} e^{-at} u(t) + C$$

(ii) $g(t) = e^{-|t|}$

$$|t| = -t < 0 < t$$

(5)

$$(i) \quad g_1(t) = e^{-at} u(t)$$

$$G_1(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-at - j\omega t} dt$$

$$= \int_0^{\infty} e^{-t(a + j\omega)} dt$$

$$= \left. \frac{e^{-t(a + j\omega)}}{-(a + j\omega)} \right|_0^{\infty}$$

$$\frac{1}{x} = \frac{1}{|x| \angle \theta} = \frac{1}{|x|} (\angle \theta)^{-1}$$

$$= \frac{1}{\sqrt{a^2 + \omega^2}} e^{-j\theta_{g_1}(\omega)}$$

$$= -\frac{1}{a + j\omega} \left[e^{-\infty(a + j\omega)} - e^0 \right]$$

$$= -\frac{1}{a + j\omega} [0 - 1]$$

input $\theta_{g_1}(\omega)$
 output $\theta_y(\omega)$
 Impulse $\theta_h(\omega)$

$$= \frac{1}{a + j\omega}$$

$$G_1(\omega) = |G_1(\omega)| e^{j\theta_{g_1}(\omega)}$$

$$|G_1(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}, \quad \theta = \tan^{-1}\left(\frac{\omega}{a}\right)$$

$$e^{-a|t|} = e^{-at} u(t) + e^{+at} u(-t)$$

(6)

$$g_2(t) = e^{-a|t|} \rightarrow \int_{-\infty}^0 \int_0^{\infty}$$

$$G_2(\omega) = \int_{-\infty}^0 e^{+at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \left. \frac{e^{t(a - j\omega)}}{a - j\omega} \right|_{-\infty}^0 + \left. \frac{e^{-t(a + j\omega)}}{-(a + j\omega)} \right|_0^{\infty}$$

$$= \frac{1}{a - j\omega} (e^0 - e^{-\infty}) - \frac{1}{a + j\omega} (e^{\infty} - e^0)$$

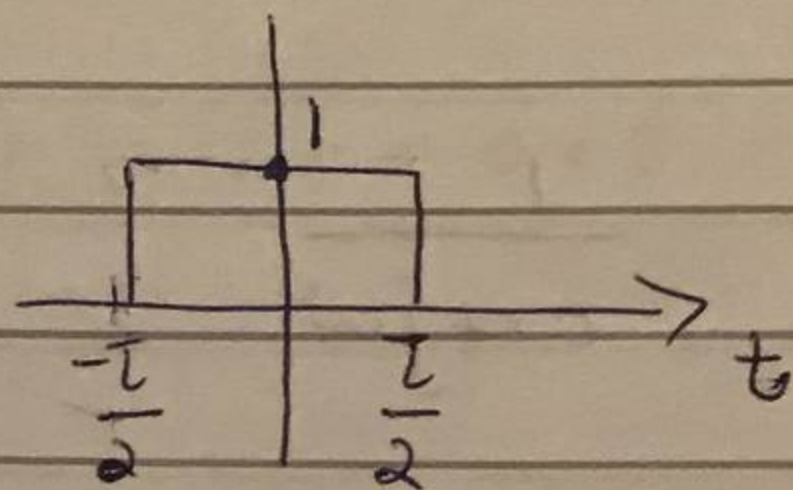
$$= \frac{1}{a - j\omega} = \frac{1}{a + j\omega} + \frac{1}{a - j\omega}$$

$$G_2(\omega) = \frac{1}{a + j\omega} + \frac{1}{a - j\omega}$$

$$= \frac{(a - j\omega)^* (a + j\omega)}{a^2 + \omega^2} = \frac{2a}{a^2 + \omega^2}$$

$$(iii) g_3(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

$$G_3(\omega) = \int_{-\tau/2}^{\tau/2} 1 \cdot e^{-j\omega t} dt$$



(7)

$$G_3(\omega) = \int_{-T/2}^{T/2} e^{-j\omega t} dt$$

$$= \frac{-1}{j\omega} e^{-j\omega t} \Big|_{-T/2}^{T/2}$$

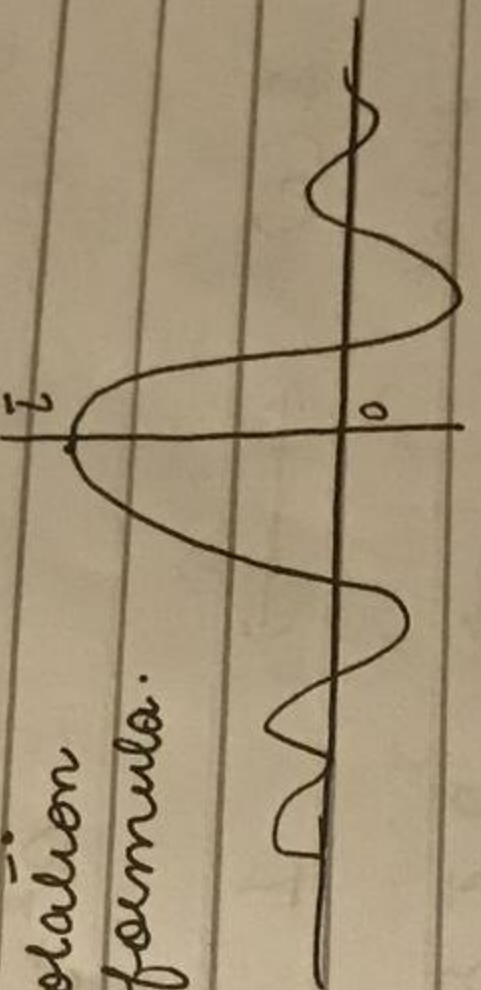
$$= \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega}$$

$$= \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j} = \sin(\omega T/2)$$

$$= \frac{2}{j\omega} \left(-e^{-j\omega T/2} + e^{j\omega T/2} \right)$$

$$= + \frac{T/2}{\omega T/2} \sin(\omega T/2)$$

filter/interpolation formula.

$$= 2 \times \frac{T}{2} \operatorname{sinc}\left(\frac{\omega T}{2}\right)$$


To find zero crossings, find $\frac{\omega T}{2} = \pm n\pi$

$$\Rightarrow \omega = \pm \frac{2n\pi}{T}$$

Compact Fourier Series:

$$\sum_{n=0}^{\infty} C_n \cos(n\omega t + \theta_n), \quad n = 0, 1, 2, \dots$$

Max. signal energy $\rightarrow$ Bandwidth approx.

$$B.W \text{ of } \sin\left(\frac{2\pi}{T}\right) = \frac{2\pi}{T} \text{ rad/s}$$

$$= \frac{1}{T} \text{ Hz}$$

Pairs:

$$\begin{aligned} e^{-at} u(t) &\longleftrightarrow \frac{1}{a+j\omega} \\ e^{at} u(-t) &\longleftrightarrow \frac{1}{a-j\omega} \\ e^{-a|t|} &\longleftrightarrow \frac{2a}{a^2 + \omega^2} \end{aligned}$$

$$\delta(t) \longleftrightarrow 1$$

$$e^{j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega - \omega_0)$$

$$\begin{aligned} \cos(\omega_0 t) &\longleftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \\ \sin(\omega_0 t) &\longleftrightarrow j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] \end{aligned}$$

$$u(t) \longleftrightarrow \pi \delta(\omega) + \frac{1}{j\omega}$$

$$\begin{aligned} \operatorname{sgn}(t) &= \frac{j\omega}{2} \\ \Delta\left(\frac{t}{T}\right) &= \frac{2}{T} \operatorname{sinc}^2\left(\frac{\omega T}{4}\right) \end{aligned}$$

(9)

Duality/Symmetry:

$$g(t) \longleftrightarrow G(\omega)$$

$$G(t) \longleftrightarrow 2\pi g(-\omega)$$

e.g.

$$g(t) = \text{sinc}\left(t \frac{\tau}{4}\right)$$

$$\text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\frac{1}{\tau} \text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\text{sinc}\left(t \frac{\tau}{2}\right) \longleftrightarrow \frac{2\tau}{1} \cdot \frac{1}{\tau} \text{rect}\left(-\frac{\omega}{\frac{1}{\tau}}\right)$$

$$\frac{1}{\tau} \text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\longleftrightarrow \frac{4\tau}{t} \text{rect}\left(\frac{2\omega}{\tau}\right)$$

Scaling Property:

$$g(at) = \frac{1}{|a|} G\left(\frac{\omega}{a}\right)$$

convolution with impulse $\xrightarrow{\text{result}}$ same signal

(10)

Time Shifting:

$$g(t) \Leftrightarrow G(\omega)$$

$$g(t - t_0) \Leftrightarrow G(\omega) e^{-j\omega t_0} \quad \text{phase shift}$$

In polar form:

$$\Leftrightarrow |G(\omega)| e^{j\theta_g(\omega)}$$

$$\text{e.g. } e^{-a|t-t_0|} \Leftrightarrow \frac{2a}{a^2 + \omega^2} \cdot e^{-j\omega t_0}$$

$$\frac{2a}{a^2 + \omega^2} e^{-j\omega t_0}$$

Frequency Shifting:

$$g(t) e^{+j\omega_0 t} \Leftrightarrow G(\omega - \omega_0)$$

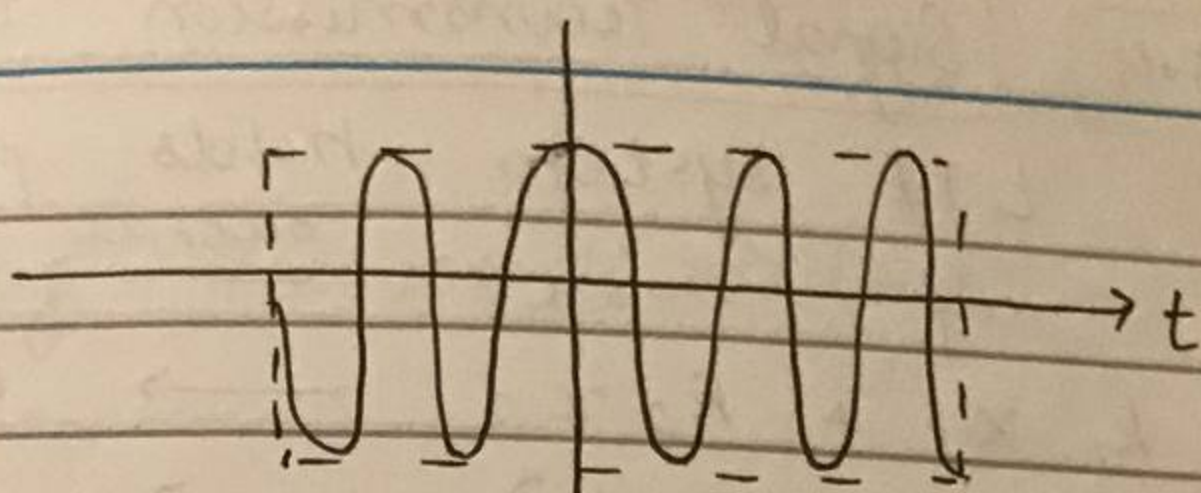
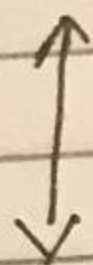
$$g(t) e^{-j\omega_0 t} \Leftrightarrow G(\omega + \omega_0)$$

$$g(t) \cos(\omega_0 t) \Leftrightarrow \frac{1}{2} [G(\omega + \omega_0) + G(\omega - \omega_0)]$$

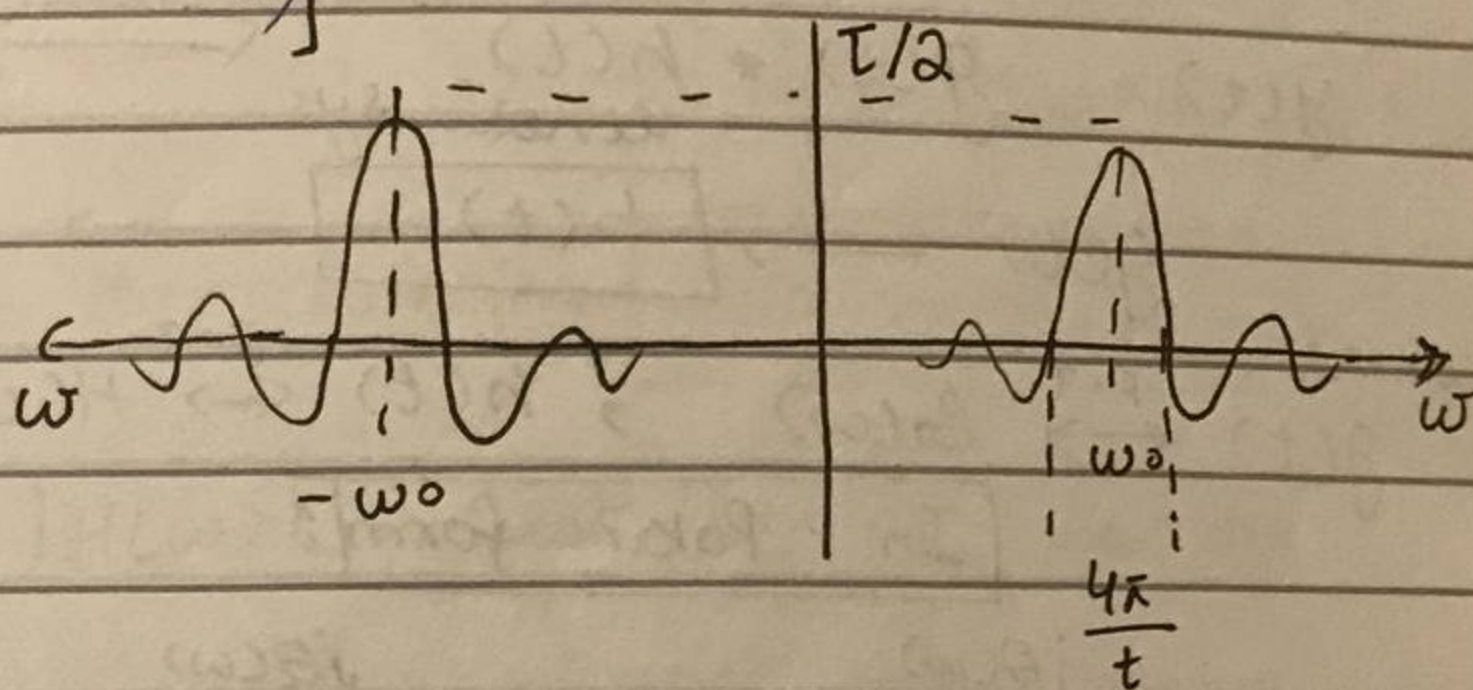
$$G(\omega) jk \sin(\omega T) \Leftrightarrow \frac{k}{2} [g(t+T) - g(t-T)]$$

(ii)

$$\text{rect}\left(\frac{t}{\tau}\right) \cos(\omega_0 t)$$



$$\frac{\tau}{2} \left[\sin\left(\frac{\tau}{2}(\omega - \omega_0)\right) + \sin\left(\frac{\tau}{2}(\omega + \omega_0)\right) \right]$$



Lec #03:

22-08-2017

2 B Hz

$$BW = \frac{4\pi}{\tau} \frac{\text{rad}}{\text{s}}$$

$$= \frac{2}{\tau} \text{ Hz}$$

$$g_1(t) * g_2(t) \leftrightarrow G_1(\omega) G_2(\omega)$$

$$g_1(t) \cdot g_2(t) \leftrightarrow \frac{1}{2\pi} G_1(\omega) * G_2(\omega)$$

$$g^n(t) \quad \text{Bandwidth} = n B(\text{Hz})$$

3.4 Signal Transmission through a linear system

LTI system holds principle of superposition

$$\begin{aligned}
 & \text{If } x(t) \xrightarrow{\text{output}} y(t) \\
 & k_1 x_1 + k_2 x_2 \longrightarrow k_1 y_1 + k_2 y_2 \quad \text{Need!!} \\
 & x(t - t_0) \longrightarrow y(t - t_0)
 \end{aligned}$$

$$y(t) = g(t) * h(t) \quad \xleftrightarrow{\text{F.T}} \quad Y(\omega) = G(\omega) H(\omega)$$

linear sys

$$g(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$

where

$$g(t) \xleftrightarrow{\text{F.T}} G(\omega), \quad h(t) \xleftrightarrow{\text{F.T}} H(\omega), \quad y(t) \xleftrightarrow{\text{F.T}} Y(\omega)$$

In Polar form:

$$|Y(\omega)| e^{j\theta_y(\omega)} = |G(\omega)| e^{j\theta_g(\omega)} |H(\omega)| e^{j\theta_h(\omega)}$$

$$|Y(\omega)| e^{j\theta_y(\omega)} = |G(\omega)| |H(\omega)| e^{j(\theta_g(\omega) + \theta_h(\omega))}$$

$$\Rightarrow |Y(\omega)| = |G(\omega)| |H(\omega)| \quad \&$$

$$\theta_y(\omega) = \theta_g(\omega) + \theta_h(\omega)$$

Distorted Signal: If has frequency component ko diff. attenuation milay or har kisi ko diff. phase shift/delay milay then signal is called distorted.

$$y(t) = k g(t - t_d)$$

(1) ↑ fixed amount of attenuation for whole signal (for every freq. comp)
 (2) same delay for every frequency comp. Distortionless signal.

In polar form:

$$Y(\omega) = k G(\omega) e^{-j\omega t_d}$$

compare the above eq. by $Y(\omega) = H(\omega) G(\omega)$
to find $H(\omega)$

$$\Rightarrow H(\omega) = k e^{-j\omega t_d}$$

$$|H(\omega)| = k$$

$$\theta_h(\omega) = -\omega t_d$$

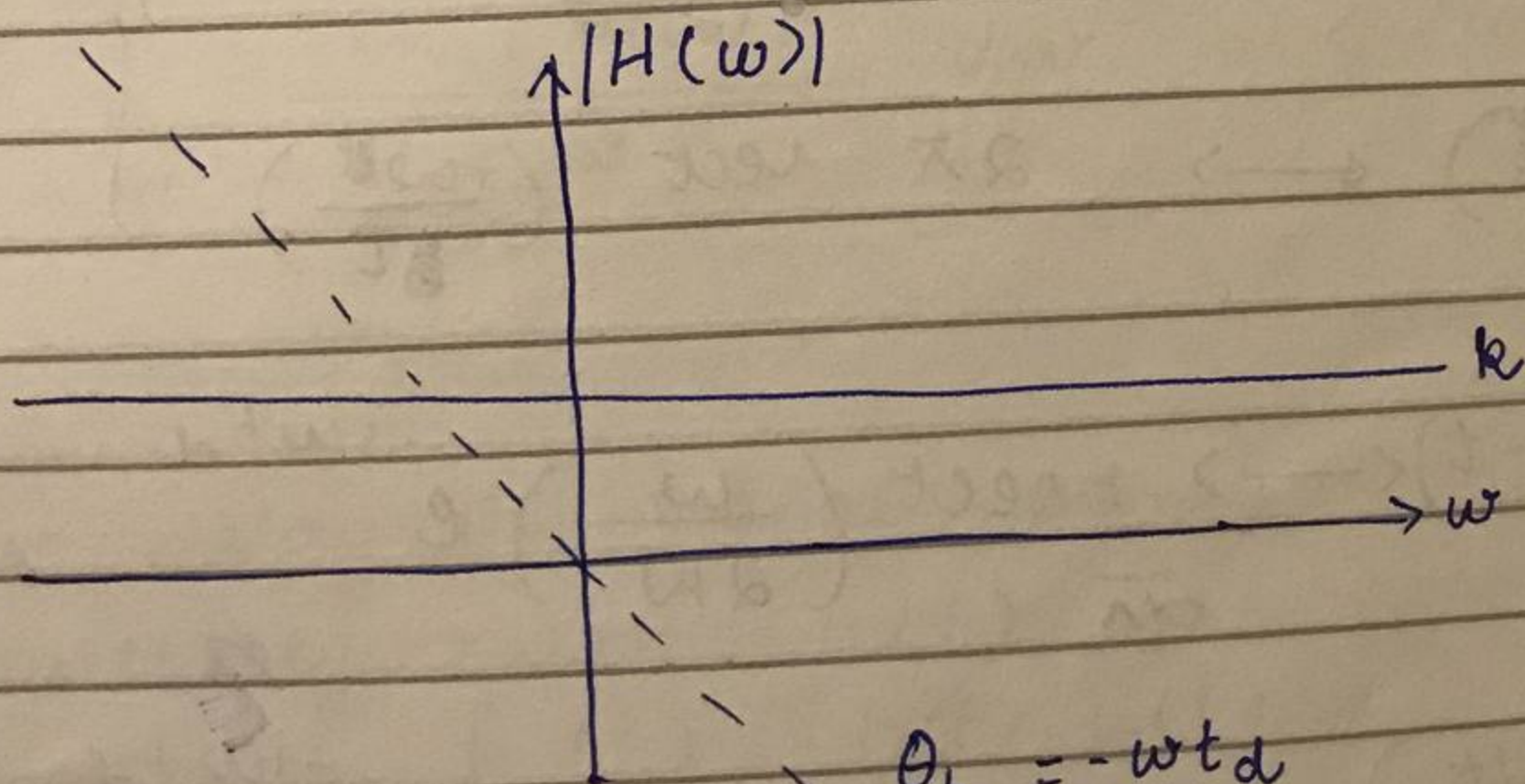
← since θ is a function of ω so θ will change with ω

Conditions for Distortionless ~~Bandwidth~~ Transmission:

① → Amplitude response $|H(\omega)|$ must be a constt.
i.e. $|H(\omega)| = k$

② → Phase response $\theta_h(\omega)$ must be a linear func. of ω i.e. $\theta_h(\omega) = -\omega t_d$

① & ② are only for the frequency bands in which we are interested.
↑ "BAND OF INTEREST"



$$\theta_h = -\omega t_d$$

slope ?

$$\boxed{-\frac{d\theta}{d\omega} = t_d} \rightarrow \text{constant.}$$

time delay (sec)

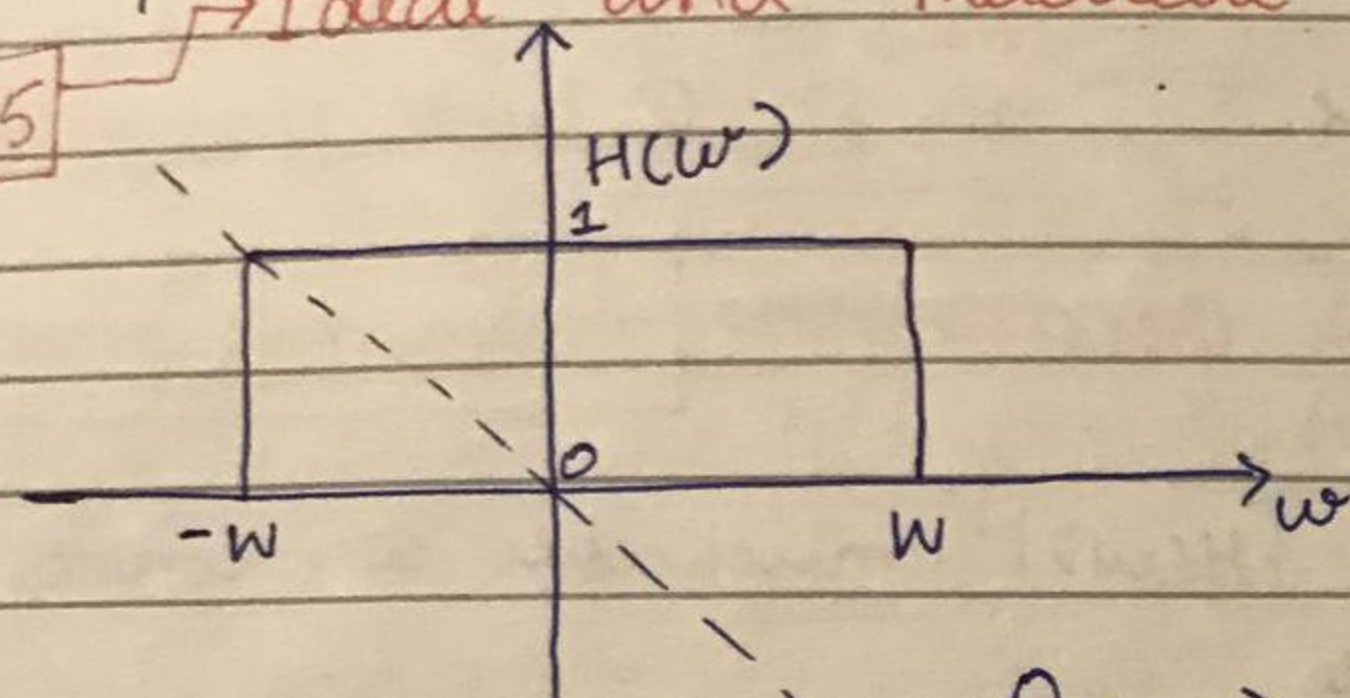
$$\theta = -\omega t_d \quad \text{constant}$$

↑
phase shift

Phase → should be a func. of "ω"
Resp.

→ Ideal and Practical filters

3.5



$$y(t) = g(t - t_d)$$

$$|H(\omega)| = 1 \quad -W < \omega < W$$

$$H(\omega) = \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d} \quad (\text{polar form})$$

$$h(t) = ?$$

$$\text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\tau \text{sinc}\left(\frac{T\omega}{2}\right) \longleftrightarrow 2\pi \text{rect}\left(\frac{\omega}{2\pi T}\right)$$

$$\frac{2W}{2\pi} \text{sinc}\left(\frac{W \cdot t}{\pi}\right) \longleftrightarrow \frac{1}{2\pi} \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d}$$

$$\frac{2W}{2\pi} \text{sinc}\left(\frac{Wt}{\pi}\right) \longleftrightarrow \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d}$$

for zero crossings:

$$Wt = \pm n\pi$$

$$t = \pm \frac{n\pi}{W}$$

(15)

$$\frac{W}{\pi} \text{sinc}(W(t - t_d)) \longleftrightarrow \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d}$$

Graphically:

Check in ① time Domain
(causal/non-causal)

Non-causal signal

$$h(t) \neq 0 \text{ for } t < 0$$

causal
• Ideal $h(t)$ system because its appearing before '0'.

OR
↓

② In Frequency Domain
(check if the signal is physically ~~possible~~ or not)
realisable

PALEY - WIENER CRITERION:

$$\int_{-\infty}^{\infty} \frac{\ln |H(\omega)|}{1 + \omega^2} d\omega < \infty \quad (\text{Realizable if integration is } < \infty)$$

First delay the signal then

multiply with $u(t)$

$$\hat{h}(t) = h(t) u(t)$$

↑ better approximation of $h(t)$
because we have delayed t_d
by 4 times i.e. $t_d = 4 t_d$

Q: $f = 20 \text{ kHz}$
 $t_d = 0.1 \text{ msec}$ is enough for the above req.
 How is it enough?
 Sol: $\omega = 2\pi f = 2\pi \times 20 \text{ kHz}$
 $= 40\pi \text{ kHz}$

increasing delay by $4t_d$

$$4t_d = 4 \times \frac{1}{\omega} = \frac{4}{40\pi \times 10^3} = 4 \times 10^{-3} \text{ sec}$$

$= 0.1 \times 10^{-3} \text{ sec}$ Hence proved.

Example #3.16

variation in amplitude response.
 time delay is tolerable.

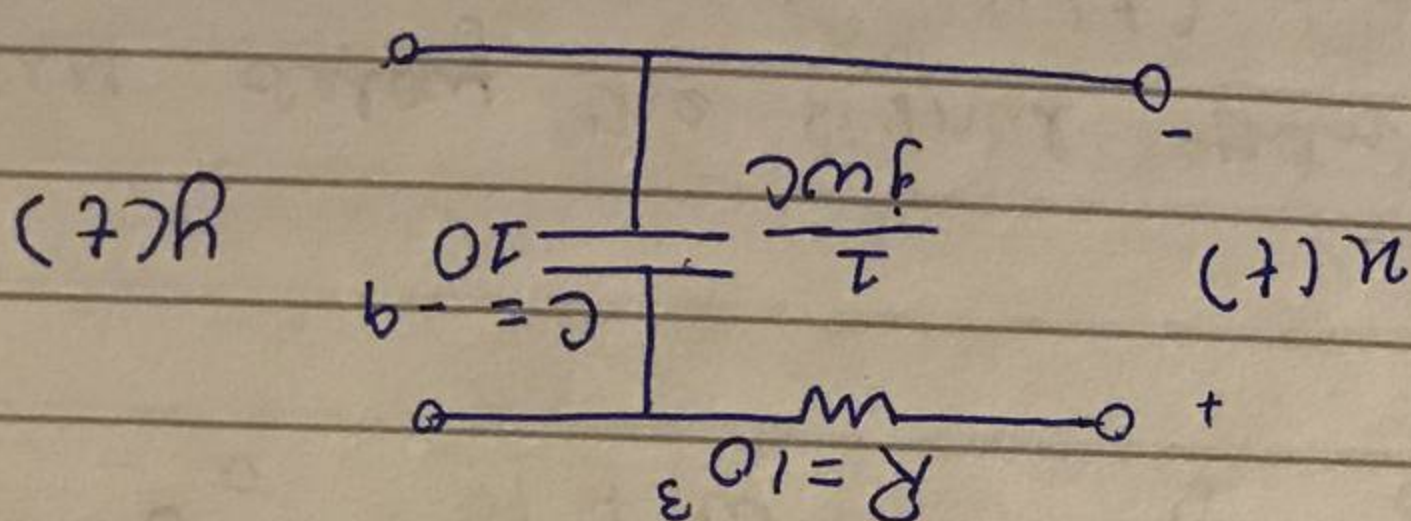
$$\frac{Y(\omega)}{X(\omega)} = \frac{1/j\omega C}{R + 1/j\omega C}$$

$$H(\omega) = \frac{1/R}{1 + j\omega RC}$$

put $a = \frac{1}{RC} = 10^6$

$$\Rightarrow \frac{a + j\omega}{a} = H(\omega)$$

$$\frac{a + jb}{a + jb} = \frac{1}{1 + j\omega RC}$$



$$\frac{d}{dx} \left(\tan^{-1} \left(\frac{x}{a} \right) \right) = \frac{a}{a^2 + x^2}$$

(17)

e.g. $\frac{0.008 \times 10^{-3}}{500 \times 10^3}$

$$|H(\omega)| = \sqrt{\frac{a^2}{a^2 + \omega^2}} = \frac{a}{\sqrt{a^2 + \omega^2}} \approx \frac{1}{\text{provided } \omega \ll a}$$

$$\theta_h(\omega) = -\tan^{-1} \left(\frac{\omega}{a} \right)$$

$$\boxed{\omega \ll a}$$

$$\approx -\frac{\omega}{a} \quad (\text{using calculus})$$

$$t_d(\omega) = \frac{-d\theta_h}{d\omega} = -\frac{d}{d\omega} \left(-\tan^{-1} \left(\frac{\omega}{a} \right) \right)$$

$$= \frac{a}{a^2 + \omega^2} \approx \frac{1}{a} = 10^{-6} \text{ s}$$

Gain of L.P.F = $\left(\frac{1}{\sqrt{2}} \right)^{(n)}$ order of L.P.F provided $\omega \ll a$

t_d has a maximum value at $\omega = 0$ of $\frac{1}{a}$
 $|H(\omega)|$ " " " " " " " "

$$y(t) = k \underset{\text{amplitude}}{g(t - t_d)} \quad \text{time delay}$$

~~amplitude~~

3.6 Signal Distortion over a communication channel

① Linear Distortion:

- a. $|H(\omega)| \neq k \rightarrow$ Amp. distortion
- b. $\theta_h(\omega) \neq -\omega t_d \rightarrow$ Phase "

If a exists or b exists or both exist then linear distortion exists

(18)

Because of linear distortion

effects

Pulse spreading / Dispersion

TDM Signal

↳ Doesn't effect FDM.

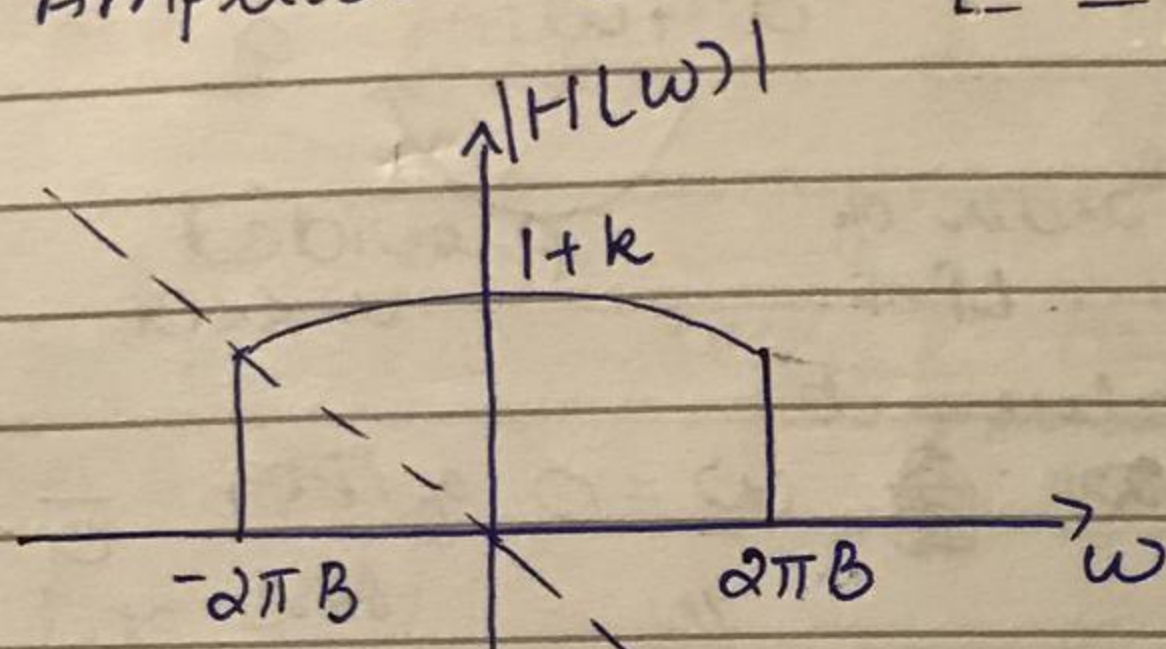
Time Domain
Multiplexing

For Example:

$$H(\omega) = \begin{cases} (1 + k \cos T\omega) e^{-j\omega t_d} & |\omega| \leq 2\pi B \\ 0 & |\omega| > 2\pi B \end{cases}$$

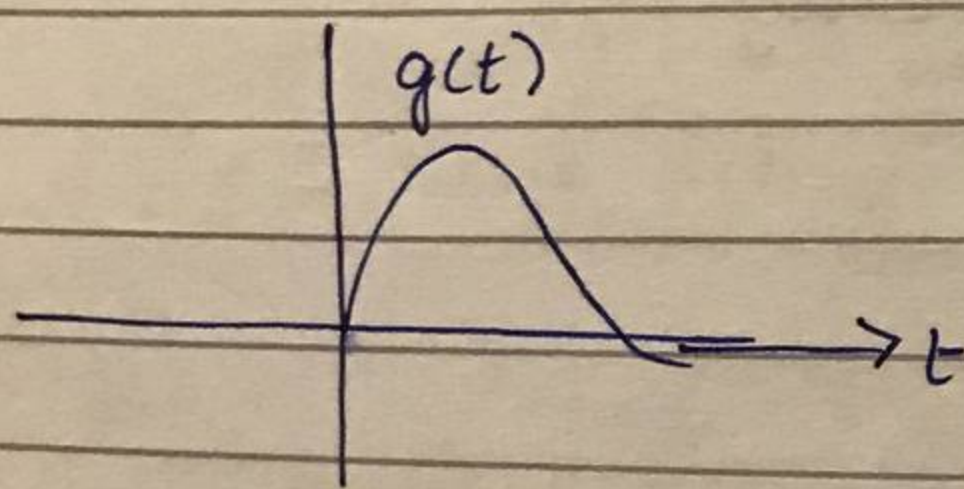
Amplitude Distortion

Pulse Dispersion



$$Y(\omega) = G(\omega) \cdot$$

$$(1 + k \cos T\omega) e^{-j\omega t_d}$$



-ve slope

(that is why Π and
IV - Quad)

$$-j\omega t_d$$

$$g(t - t_d)$$

$$Y(\omega) = G(\omega) e^{-j\omega t_d} + k G(\omega) \cos(T\omega) e^{-j\omega t_d}$$

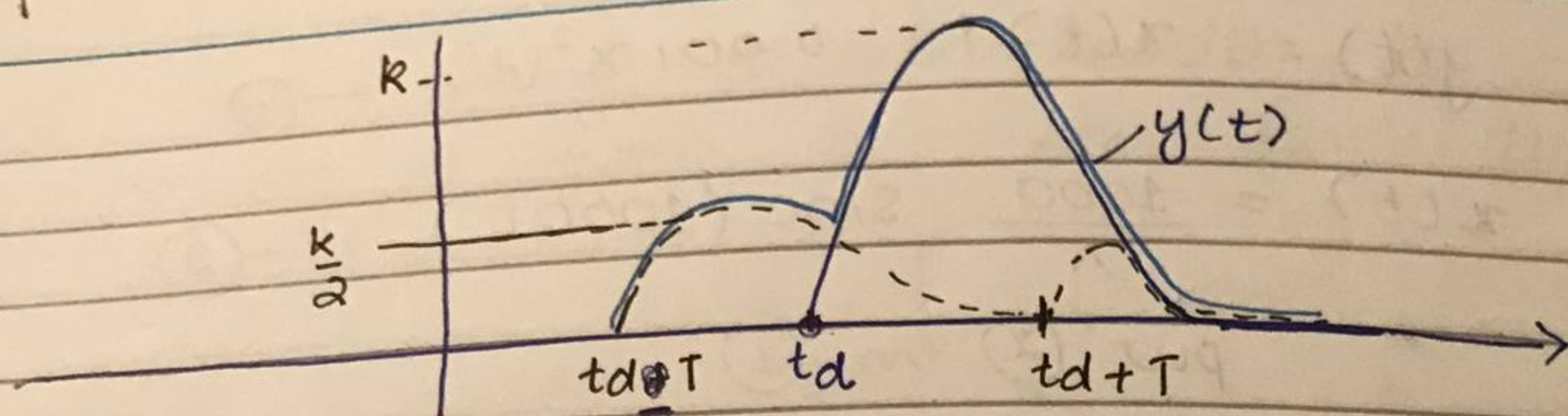
$$= g(t - t_d) + k \left[\frac{1}{2} (g(t + T) + g(t - T)) \right] e^{-j\omega t_d}$$

$$g(t - T)] e^{-j\omega t_d}$$

$$= g(t - t_d) + \frac{k}{2} \left[(g(t + T - t_d) + g(t - (T + t_d))) \right]$$

$$(2-216(2)6 \int)$$

(19)



- By adding the two signals (dark blue + black)

Non-Linear channel

Distortions caused by Channel Non-linearities

→ Spectral spreading / dispersion

$$y = f(g)$$

$$B \cdot W = k \text{ BHz}$$

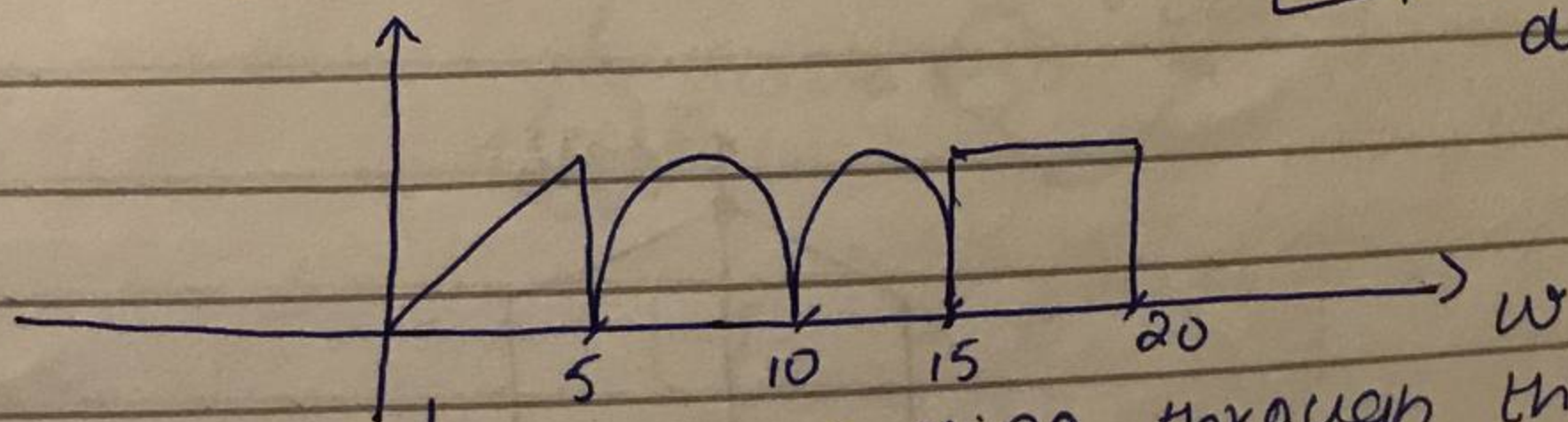
Maclaurin series:

$$y(t) = a_0 + a_1 g'(t) + \dots + a_k g^k(t) + \dots$$

$$+ a_2 g_0^2(t) \rightarrow B \cdot W = 2B$$

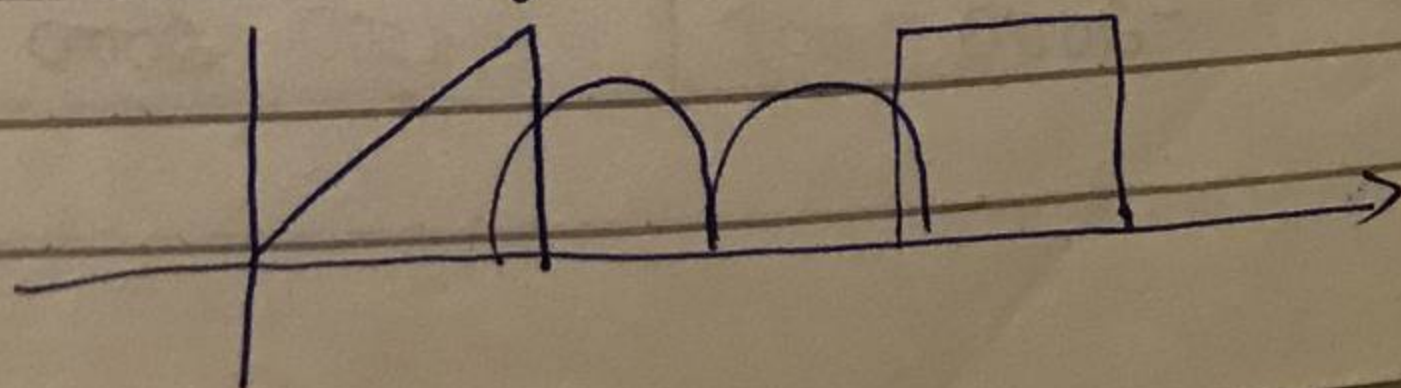
→ Interference problem in FDM.

e.g.



frequency division multiplexing

↓ after passing through the system



(20)

Q. $y(t) = x(t) + 0.001x^2(t)$ — (1)

$x(t) = \frac{1000}{\pi} \text{sinc}(1000t)$ — (2)

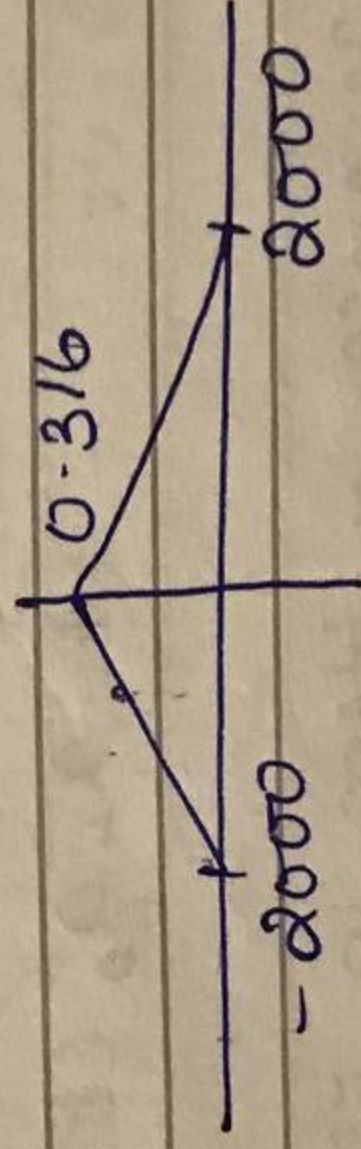
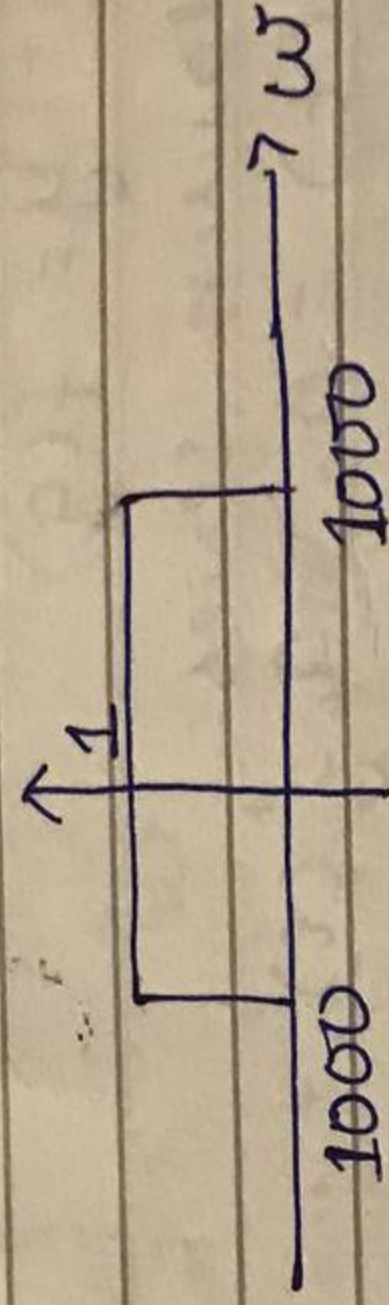
Sol.

put (2) in (1)

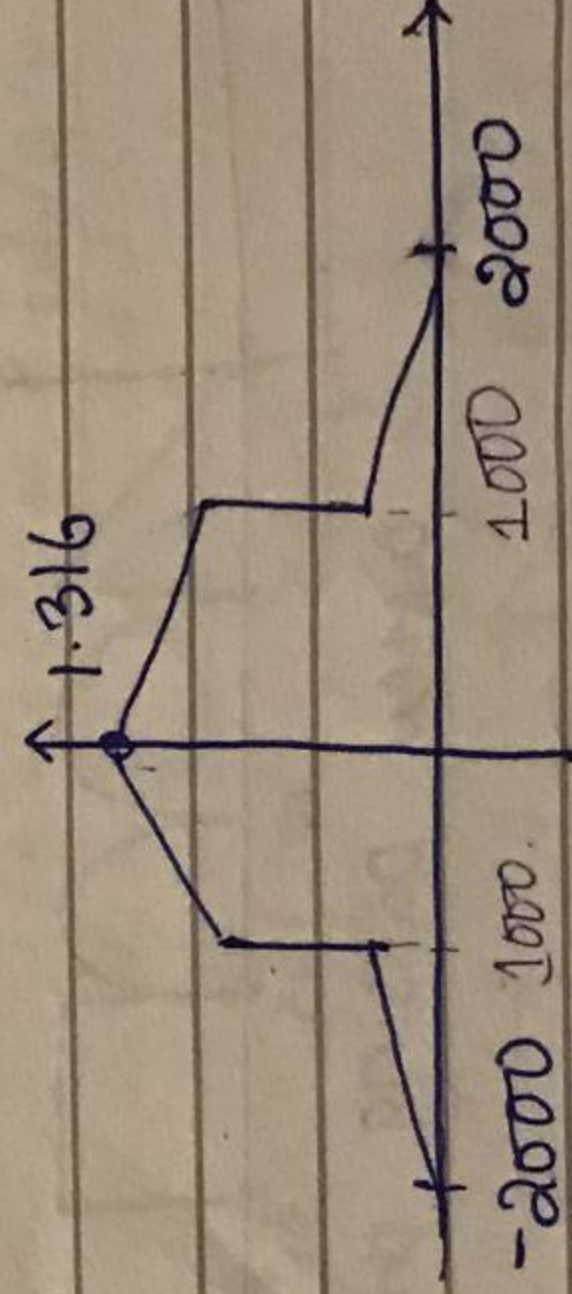
$y(t) = \frac{1000}{\pi} \text{sinc}(1000t) + 0.001 \left(\frac{1000}{\pi} \text{sinc}(1000t) \right)^2$

After calculations:

$Y(\omega) = \underbrace{\text{rect}\left(\frac{\omega}{2000}\right)}_{\text{Desired signal}} + \underbrace{0.316 \Delta\left(\frac{\omega}{4000}\right)}_{\text{Distorted signal (undesired)}} \quad (3)$

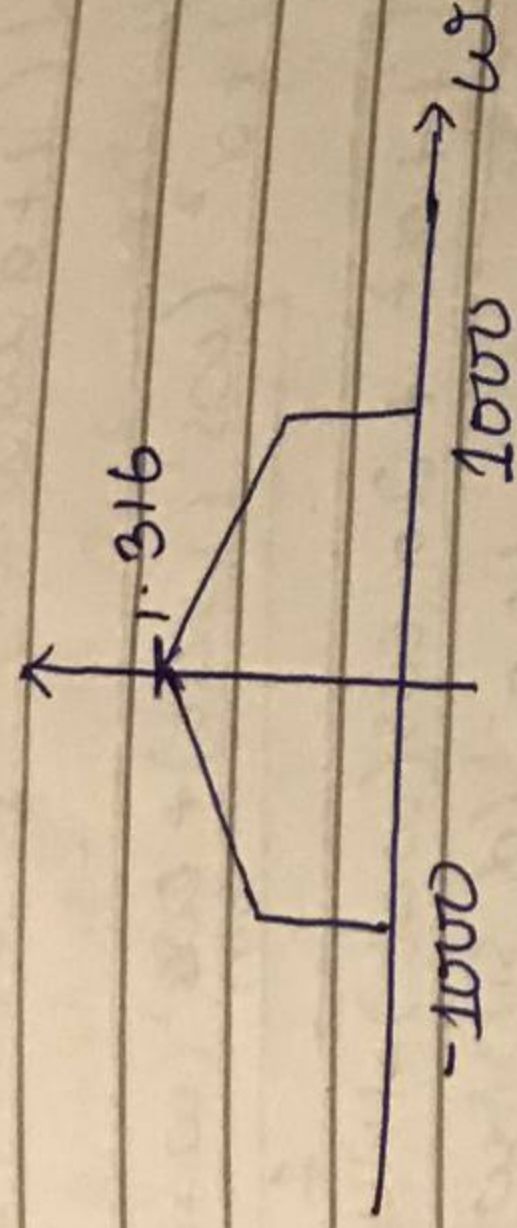


eq. no (3)



(21)

we can change its Bandwidth by passing it from a Low Pass Filter.



It can't be recovered in " ω - Domain" but in Time - Domain signal it can be recovered. (TDM)

$\Rightarrow$ Distortion caused by Multipath effect:

Multiple signals at receiver end due to mis-matching of impedances.

Differences:

Attenuation $|H(\omega)| = k$ Delay $\theta_h(\omega) = -\omega t_d$

$$H(\omega) = e^{-j\omega t_d} + \alpha e^{-j\omega(t_d + \Delta t)}$$

$$= e^{-j\omega t_d} [1 + \alpha e^{-j\omega \Delta t}]$$

$$= e^{-j\omega t_d} [1 + \alpha \cos(\omega \Delta t) - j\alpha \sin(\omega \Delta t)] \quad \text{--- (1)}$$

(22)

$$\begin{aligned}
 |H(\omega)| &= \sqrt{(\text{real part})^2 + (\text{gm. part})^2} \\
 &= \sqrt{(1 + \alpha \cos(\omega \Delta t))^2 + (\alpha \sin(\omega \Delta t))^2} \\
 &= \sqrt{1 + \alpha^2 (\underbrace{\cos^2(\omega \Delta t) + \sin^2(\omega \Delta t)}_1) + 2\alpha \cos(\omega \Delta t)} \\
 &= \sqrt{1 + \alpha^2 + 2\alpha \cos(\omega \Delta t)} \rightarrow |H(\omega)| \text{ amp.} \\
 &= e^{-j [\omega \Delta t + \tan^{-1} \left(\frac{\alpha \sin(\omega \Delta t)}{1 + \alpha \cos(\omega \Delta t)} \right)]} \quad \begin{array}{l} \text{from } \textcircled{1} \\ \text{phase} \end{array} \\
 \theta_h(\omega) &= e^{-j \theta_h(\omega)}
 \end{aligned}$$

Conclusion:

Phase and amp. dispersion in the above example. (Linear Distortion)
 $\Rightarrow$ Pulse Dispersion occurred

duration $\Rightarrow t \rightarrow \infty$
finite signal $\Rightarrow g(t) \rightarrow 0$

3.7 Signal Energy and energy spectral density:

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt, \text{ area under } |g(t)|^2$$

Parseval's Theorem:

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} g(t) \cdot g^*(t) dt$$

$$= \int_{-\infty}^{\infty} g(t) \left[\int_{-\infty}^{\infty} \frac{1}{2\pi} G^*(\omega) e^{-j\omega t} d\omega \right] dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) \left[\int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt \right] d\omega$$

"Energy signals have 0" power.

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) G(\omega) d\omega$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

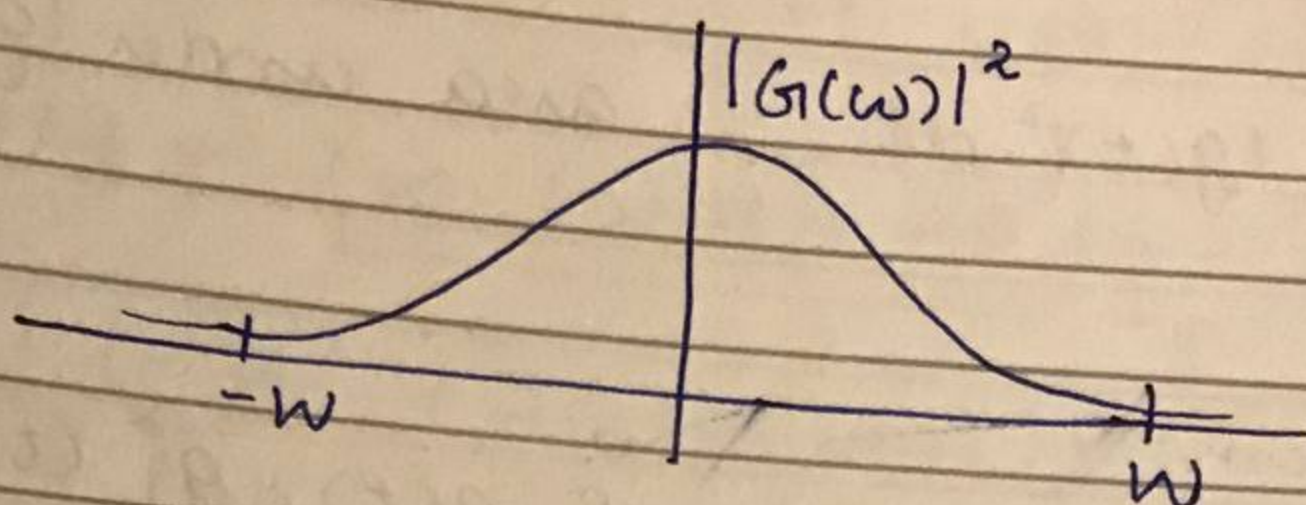
Because of symmetry:

$$E_g = \int_{-\infty}^{\infty} \frac{1}{2\pi} |G(\omega)|^2 d\omega$$

spectral density = E energy signal density = ESD
 $|G(\omega)|^2 = |G(\omega)|^2$

$\Rightarrow E_g = \frac{1}{2\pi} \times$ Area under ESD signal distribution in spectral.

Bandwidth containing max. energy of the signal $\Rightarrow$ Essential Bandwidth



$$E_g = \frac{1}{\pi} \int_0^W |G(\omega)|^2 d\omega$$

(i) 98% energy in $-W$ to W ,
find Bandwidth?

(ii) Given bandwidth,
find signal energy?

$$G(\omega) \longrightarrow \boxed{H(\omega)} \longrightarrow Y(\omega)$$

$$Y(\omega) = G(\omega) H(\omega)$$

$$|Y(\omega)|^2 = |G(\omega)|^2 |H(\omega)|^2$$

$$\Psi_y(\omega) = |H(\omega)|^2 \Psi_g(\omega)$$

ESD of output signal:

$$E_y = \int_{-\infty}^{\infty} \Psi_y(\omega) d\omega$$

$$\Rightarrow E_y = \int_{-\infty}^{\infty} |H(\omega)|^2 \Psi_g(\omega) d\omega$$

Energy of modulated signal:

$$E_{\phi} = \frac{1}{2} E_g$$

$$g(t) \cos \omega_0 t \Leftrightarrow \frac{1}{2} [G(\omega + \omega_0) + G(\omega - \omega_0)]$$

Example:

Verify Parseval's Theorem. $g(t) = e^{-at} u(t)$, $a > 0$

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

In Time Domain

$$= \int_{-\infty}^{\infty} (e^{-at})^2 dt$$

$$= \int_{-\infty}^{\infty} e^{-2at} dt$$

$$= \frac{-1}{2a} \left| e^{-2at} \right|_{-\infty}^{\infty}$$

$$= \frac{-1}{2a} [e^{-\infty} - e^{\infty}] = \frac{1}{2a} \quad - (i)$$

In Frequency Domain

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

$$G(\omega) = \frac{1}{a + j\omega}$$

$$ESD = \frac{1}{a^2 + \omega^2}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{a^2 + \omega^2} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{1}{a} \tan^{-1} \left(\frac{\omega}{a} \right) \right]_{-\infty}^{\infty}$$

$$= \frac{1}{2\pi a} \left[\tan^{-1}(\infty) - \tan^{-1}(-\infty) \right]$$

$$= \frac{1}{2\pi a} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{1}{2a} \quad \text{--- (ii)}$$

Hence proved using eq. (i) and (ii).
using last example.

①. $g(t) = e^{-at} u(t)$

if 95% of total signal energy is in $-W$ to W . Find the bandwidth.

i- calculate total energy.

ii- $0.95 E_g = \frac{1}{2\pi} \int_{-W}^W |G(\omega)|^2 d\omega$

$$0.95 E_g = \frac{1}{\pi} \int_0^W \frac{1}{a^2 + \omega^2} d\omega$$

$$0.95 E_g = \frac{1}{\pi} \left[\frac{1}{a} \tan^{-1} \left(\frac{\omega}{a} \right) \right]_{\omega=0}^{\omega=W}$$

$$0.95 E_g = \frac{1}{\pi a} \left[\tan^{-1} \left(\frac{W}{a} \right) - \tan^{-1} \left(\frac{0}{a} \right) \right]$$

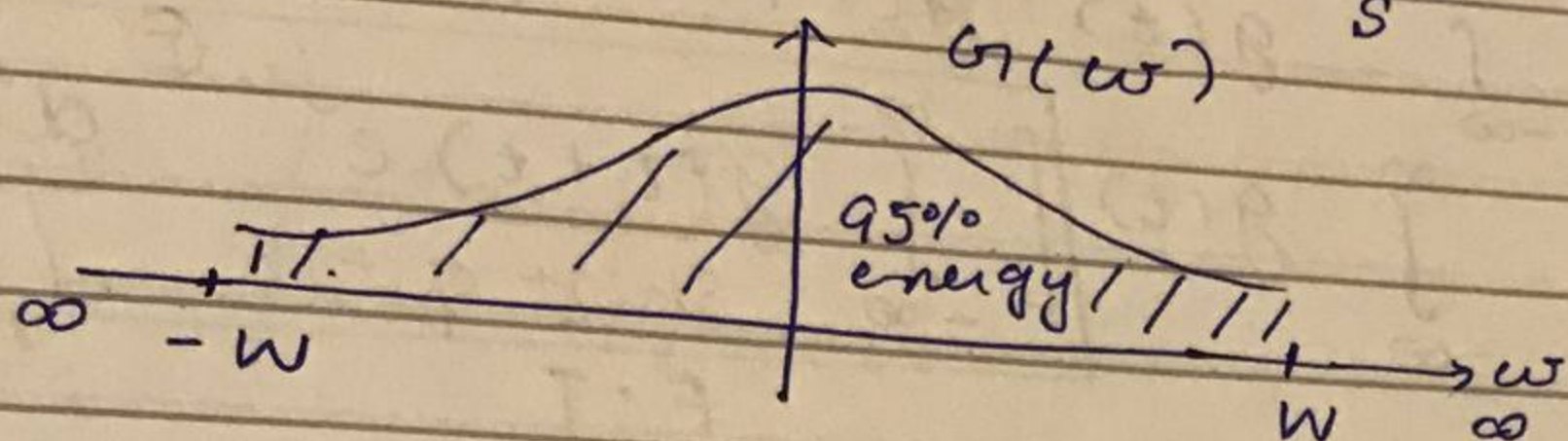
$$0.95 \cdot \frac{1}{2a} = \frac{1}{\pi a} \left[\tan^{-1} \left(\frac{W}{a} \right) - 0 \right]$$

(27)

$$1.49 = \tan^{-1}\left(\frac{W}{a}\right)$$

~~or~~ $12.70 = \frac{W}{a} \text{ rad/s}$

$$W = 12.70 a \frac{\text{rad}}{\text{s}}$$



Time auto-correlation func. and ESD

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t+\tau) dt \quad \text{--- (1)}$$

- Auto-correlation func. :
co-relation ^{b/w} ~~same~~ func and its
shifted version.

Prove (even func.)

put $x = t + \tau$ in (1)

$$= \int_{-\infty}^{\infty} g(x-\tau) g(x) dt \quad \text{--- (ii)}$$

replace x by t compare (i) and (ii)

$$= \int_{-\infty}^{\infty} g(t) \cdot g(t-\tau) dt$$

$$\tau = 0$$

$$\Psi_g(0) = \int_{-\infty}^{\infty} g(t) \cdot g(t) dt = \int_{-\infty}^{\infty} (g(t))^2 dt = E_g$$

(28)

Auto-correlation function (Italic) $\Psi_g(\tau)$ $\xleftrightarrow{F.T}$ $\Psi_g(\omega)$ ESD

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t+\tau) dt$$

$$F\{\Psi_g(\tau)\} = \int_{-\infty}^{\infty} g(t) \left[\int_{-\infty}^{\infty} \underbrace{g(\tau+t)}_{\text{Shift of } t} \underbrace{e^{-j\omega\tau}}_{\text{function of } \tau} d\tau \right] dt$$

F.T

$$= \int_{-\infty}^{\infty} g(t) G(\omega) e^{j\omega t} dt$$

$$= G(\omega) \int_{-\infty}^{\infty} g(t) e^{+j\omega t} dt$$

$$= G(\omega) G(-\omega)$$

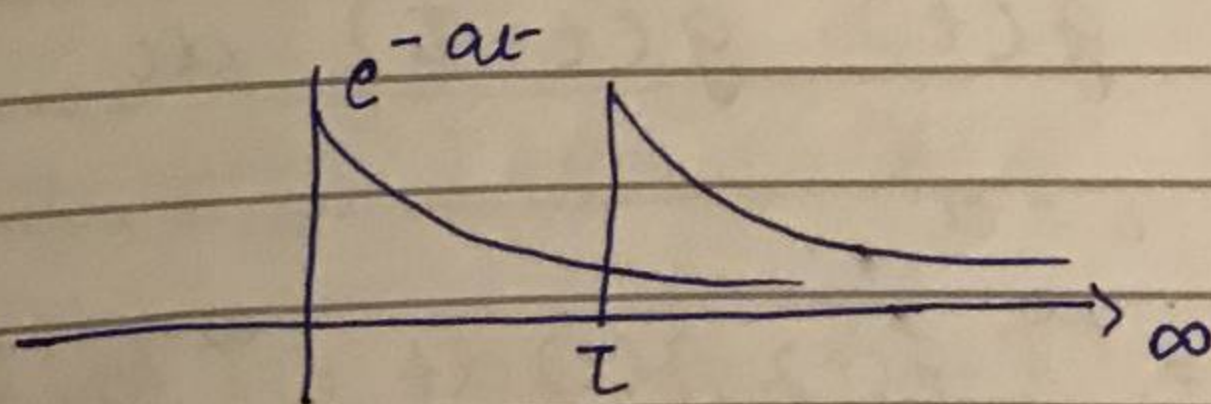
$$= |G(\omega)|^2 = \text{ESD} = \Psi_g(\omega)$$

Example # 3.22

$$g(t) = e^{-at} u(t)$$

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) \cdot g(t-\tau) dt \quad \text{right shift}$$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-a(t-\tau)} u(t-\tau) dt$$



(29)

$$= \int_{\tau}^{\infty} e^{-2at} \cdot e^{a\tau} dt$$

$$= e^{a\tau} \int_{\tau}^{\infty} e^{-2at} dt$$

$$= \frac{e^{a\tau}}{-2a} \left[e^{-2at} \right]_{\tau}^{\infty}$$

$$= \frac{e^{a\tau}}{-2a} \left[e^{-2(\infty)} - e^{-2a(\tau)} \right]$$

$$= \frac{e^{a\tau}}{+2a} (e^{2a\tau}) = \frac{e^{3a\tau}}{2a}$$

for left shift

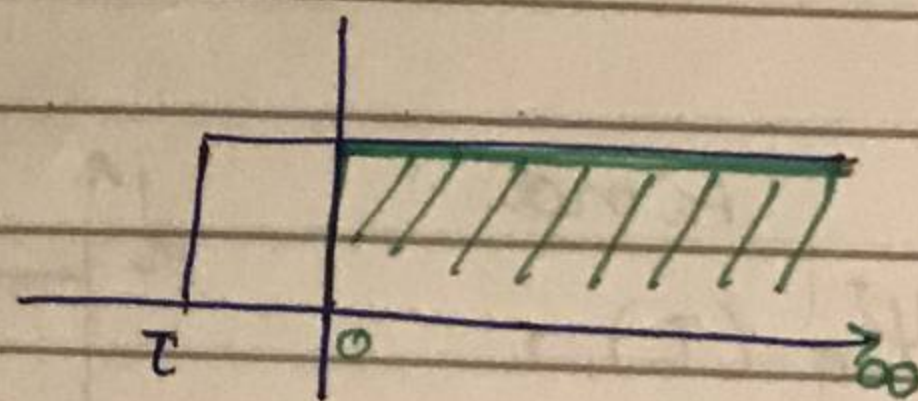
$$= \int_{-\infty}^{\infty} g(t) g(t+\tau) dt$$

$$g(t) = e^{-at} u(t)$$

$$= \int_0^{\infty} g(t) g(t+\tau) dt$$

$$= \int_0^{\infty} e^{-at} \cdot e^{-a(t+\tau)} dt$$

$$= \int_0^{\infty} e^{-a(t)} \cdot e^{-at - a\tau} dt$$



$$= e^{-at} \int_0^{\infty} e^{-2at} dt = e^{-at} \left. \frac{e^{-2at}}{-2a} \right|_0^{\infty}$$

$$= \frac{e^{-at}}{2a}$$

$$\rightarrow \Psi_g(\tau) = \Psi(-\tau)$$

Auto-correlation is an even function

i.e.

$$\Psi_g(\tau) = \frac{e^{-a|\tau|}}{2a}$$

pg # 122

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}$$

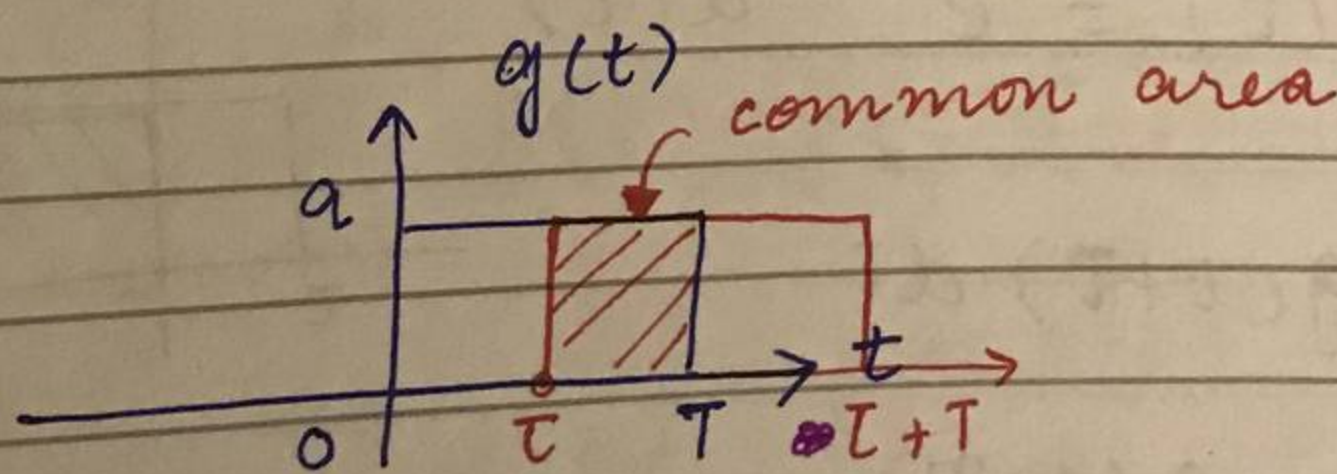
$$\frac{e^{-a|t|}}{2a} \longleftrightarrow \frac{1}{a^2 + \omega^2}$$

$$g(t) = e^{-at} u(t) \longleftrightarrow G(\omega) = \frac{1}{a + j\omega}$$

$$\Rightarrow |G(\omega)|^2 = \frac{1}{a^2 + \omega^2}$$

Q. Find $\Psi_g(\tau)$,

determine ESD from it



(31)

For right shift:

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

$$\Psi_g(\tau) = \int_{\tau}^{\tau} a^2 dt = a^2 |t| \Big|_{\tau}^{\tau}$$

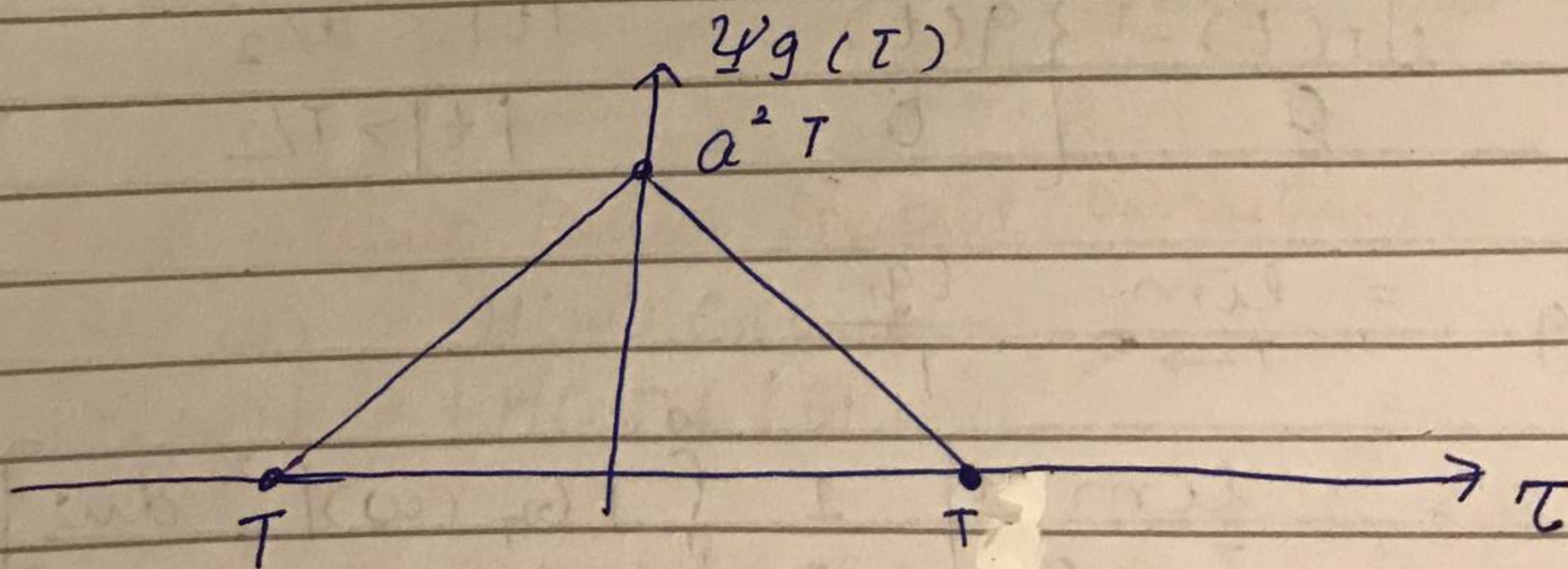
$$= a^2 [\tau - \tau]$$

Since auto-correlation is an even func. so,

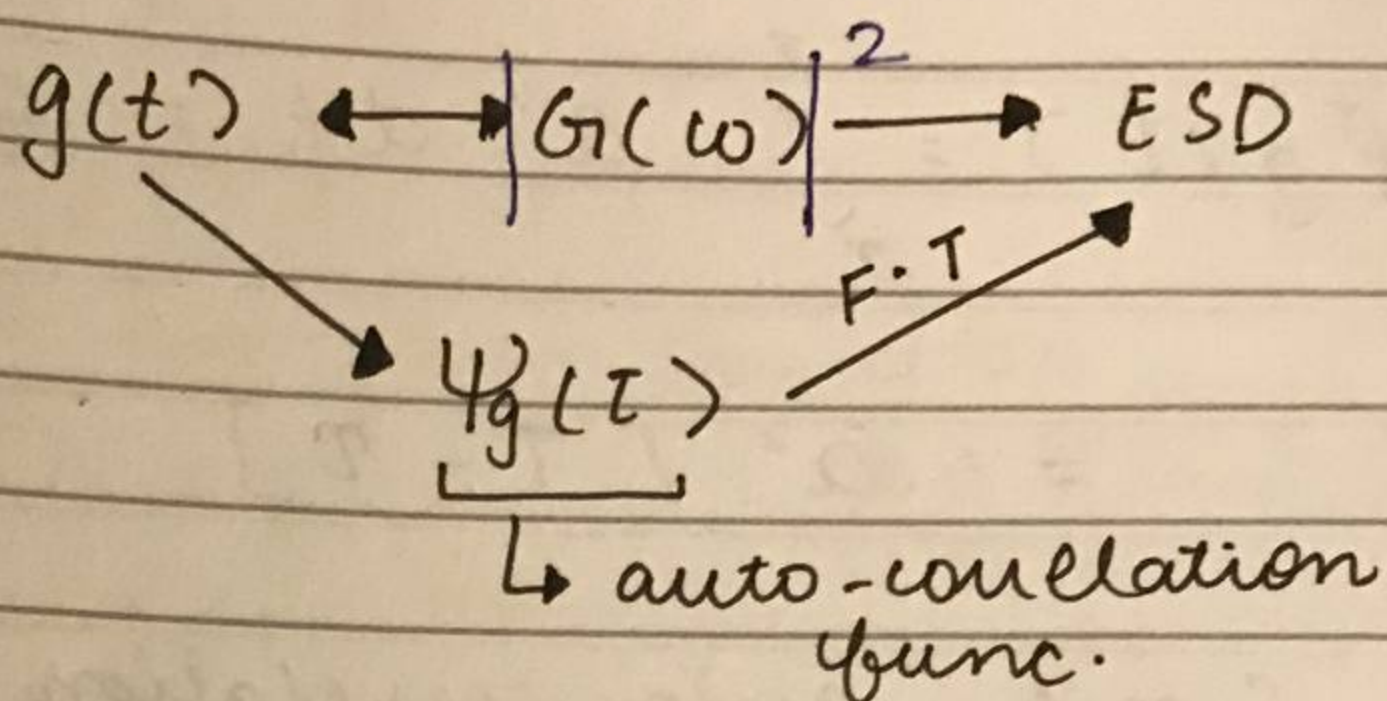
$$\Psi_g(\tau) = a^2 [\tau - |\tau|]$$

*very imp:

$$\Psi_g(\tau) = a^2 \tau \left[1 - \frac{|\tau|}{T} \right]$$



$$= a^2 T \Delta \left(\frac{t}{2T} \right) \longleftrightarrow \Psi_g(\omega) = |G(\omega)|^2$$



31-08-2017

3.8 Signal Power and Power Spectral Density

Time average of signal energy averaged over infinite time interval.

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt = \overline{g^2(t)}$$

$$g_T(t) = \begin{cases} g(t) & |t| \leq T/2 \\ 0 & |t| > T/2 \end{cases}$$

$$P_g = \lim_{T \rightarrow \infty} \frac{E_{g_T}}{T} = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} |G_T(\omega)|^2 d\omega \right]$$

As $T \rightarrow \infty$, $|G_T(\omega)|^2 \rightarrow \infty$ and the above signal will converge

Power Spectral Density

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} d\omega \quad \leftarrow S_g(\omega)$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} = \text{PSD} = \text{Power Spectral Density}$$

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_g(\omega) d\omega \rightarrow \text{Power of entire signal}$$

Conclusion:

Power is the time average of energy
 $P \cdot S \cdot D$ " " " " $E \cdot S \cdot D$

PSD and ESD are real, even

Since its an even signal, so

$$P_g = \frac{1}{2\pi} \times 2 \int_0^{\infty} S_g(\omega) d\omega$$

$$\Delta P_g = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega \rightarrow \text{Power of specific band}$$

Input and output Power Spectral Densities:

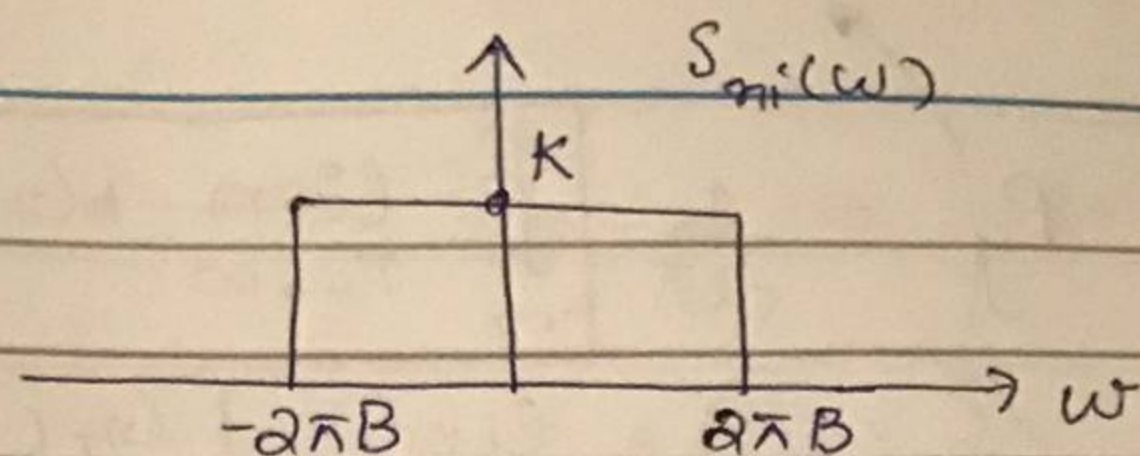
$$Y(\omega) = H(\omega) G(\omega)$$

$$|Y(\omega)| = |H(\omega)| |G(\omega)|$$

$$S_y(\omega) = |H(\omega)|^2 S_g(\omega)$$

Example # 3.24

Pg # 129

Q:

Power of output signal?

 $P = S \cdot D$?

"

?

Soln:

$$S_{ni}(\omega) = K$$

$$\therefore S_y(\omega) = |H(\omega)|^2 \cdot S_g(\omega)$$

$$S_g(\omega) = |j\omega|^2 K = \omega^2 K$$

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

$$P_y = \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} \omega^2 K d\omega$$

$$= \frac{K}{3 \times 2\pi} \left| \omega^3 \right|_{-2\pi B}^{2\pi B}$$

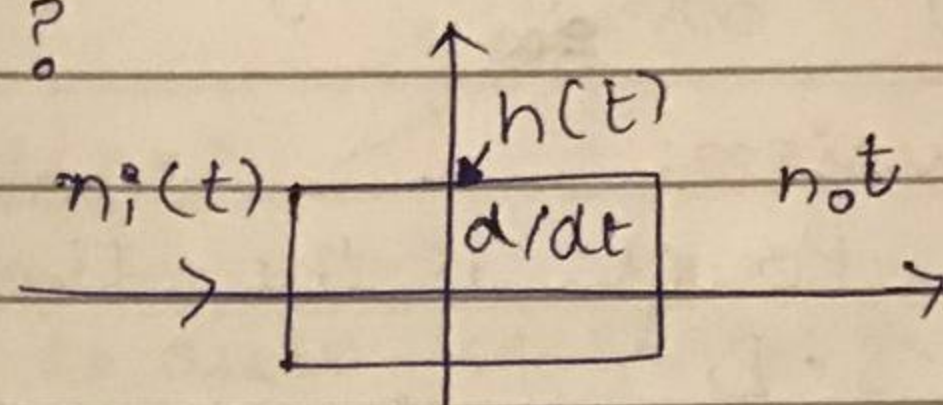
$$= \frac{K}{6\pi} \left[(2\pi B)^3 - (-2\pi B)^3 \right]$$

$$= \frac{K}{6\pi} \left(8\pi^3 B^3 - [-8\pi^3 B^3] \right)$$

$$= \frac{K}{6\pi} (8\pi^3 B^3 + 8\pi^3 B^3)$$

$$8 \times 16\pi^3 B^3 \times K$$

$$3 \times 6\pi$$



$$H(\omega) = j\omega$$

$$\begin{aligned} h(t) &= \frac{d}{dt} \\ n(t) &\leftrightarrow \frac{d}{dt} H(\omega) \\ \frac{d}{dt} &\longleftrightarrow j\omega \end{aligned}$$

(35)

$$= \frac{8}{3} \alpha^2 \beta^3 k$$

Time Auto-correlation function of Power Signal

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t) g(t+\tau) dt$$

For real $g(t)$

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g_T(t) g_T(t-\tau) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \Psi_{g_T}(\tau)$$

$$\Rightarrow R_g(\tau) = R_g(-\tau)$$

$$R_g(\tau) \xleftrightarrow{F \cdot T} \lim_{T \rightarrow \infty} \frac{\Psi_{g_T}(\omega)}{T} = S_g(\omega)$$

Advantage of Time autocorrelation:

"direct Fourier

Such power signals whose transform cannot be found, we can find their time-autocorrelation and then can find their F.T

to calculate P.S.D

if power signal
PSD (Final result)

Path 1

$$g(t) \xrightarrow{\text{Path 1}} G(\omega)$$

Path 2

Time auto-correlation

$$\Psi_{g(t)} \xleftrightarrow{F \cdot T} \Psi_{g(\omega)}$$

if energy signal $g(t)$

$$\Psi_{g(\omega)} = \text{ESD} = |G(\omega)|^2$$

made myself

(36)

Formulas: Table #3.3

Energy Signals

$$ESD = |G(\omega)|^2 = \Psi_g$$

$\Psi_g(\tau) \rightarrow$ Autocorrelation function.

$$= \int_{-\infty}^{\infty} g(t)g(t+\tau) dt$$

$$\Psi_{g(t)} \xleftrightarrow{F \leftrightarrow T} (ESD) \Psi_{g(\omega)} = |G(\omega)|^2$$

Autocorrelation is the convolution of $g(\tau)$ with $g(-\tau)$

$$\begin{aligned} E_g &= \int_{-\infty}^{\infty} g^2(t) dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |G_T(\omega)|^2 d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi_g(\omega) d\omega \end{aligned}$$

Power Signals

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} g^2(t) dt$$

$\approx$

$$P_g = \lim_{T \rightarrow \infty} \frac{E_{gT}}{T}$$

* Signal Power is the time avg. of energy.

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} d\omega$$

$\underbrace{\hspace{10em}}_{S_g(\omega)}$

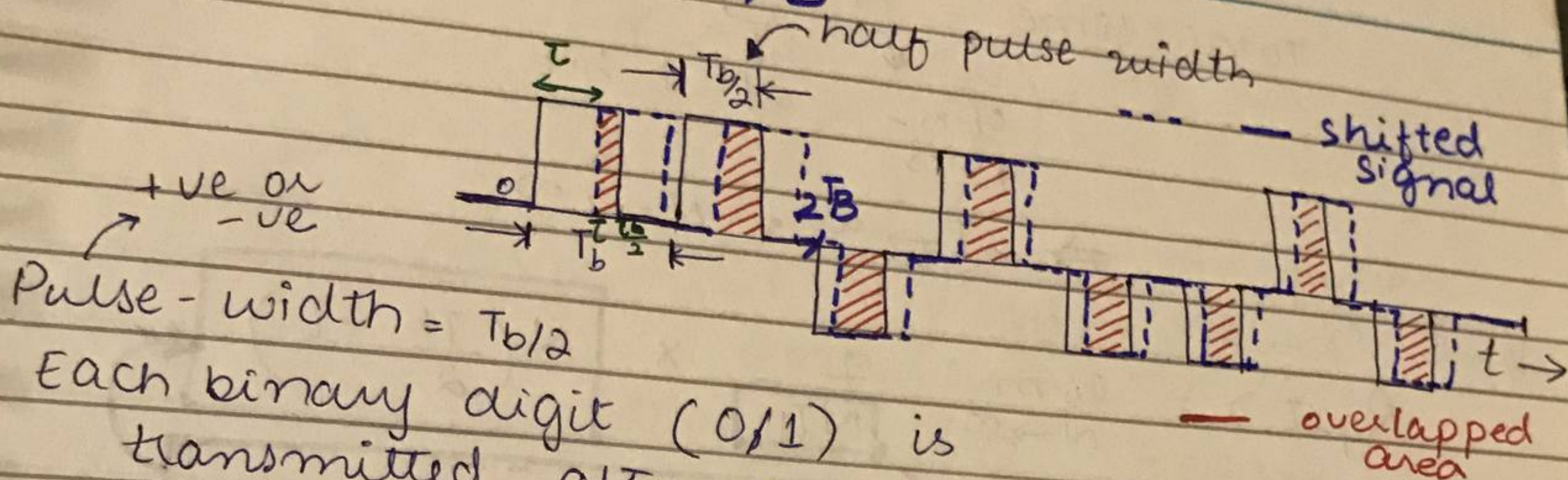
$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

$$PSD = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$$

V.IMP

(37)

Example no. 3.2, 3



Pulse-width = $T_b/2$

Each binary digit (0/1) is transmitted after T_b seconds.

1 $\rightarrow$ Positive pulse

0 $\rightarrow$ Negative pulse

Random bitstream.

Prob of 1 and 0 = 50%

Find

PSD, essential bandwidth, Time autocorrelation?

Soln.

Time auto-correlation:

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t) g(t-\tau) dt$$

(i) $\tau < \frac{T_b}{2}$ (Pulses will be overlapped by their own shifted versions, not with neighbors)

$$\int_{\tau}^{T_b/2} 1 \cdot dt$$

$$= t \Big|_{\tau}^{T_b/2}$$

$$= T_b/2 - \tau \rightarrow \text{area under } g(t) \text{ and } g(t-\tau) \text{ for one pulse}$$

For n bits:

$$N \times \left(\frac{T_b}{2} - \tau \right) = \text{Area under curve.}$$

$$\text{Total time} = \underbrace{N}_{\substack{\downarrow \\ \text{of } n\text{-} \\ \text{bits}}} \times \underbrace{T_b}_{\substack{\downarrow \\ \text{of } 1 \text{ bit}}}$$

$$\Rightarrow T \rightarrow \infty, N \rightarrow \infty$$

$$R_g(\tau) = \lim_{N \rightarrow \infty} \frac{1}{NT_b} \times N \left(\frac{T_b}{2} - \tau \right)$$

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t) g(t-\tau) dt$$

Total time

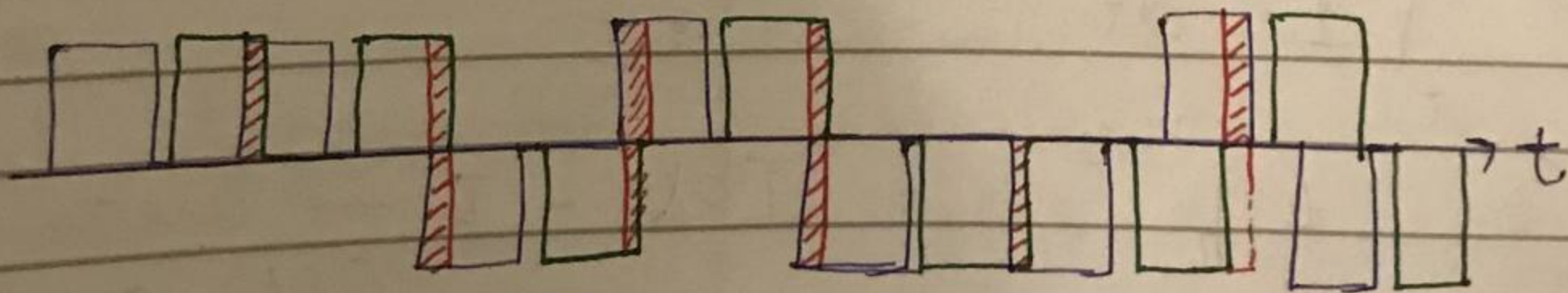
$$\Rightarrow \frac{1}{T_b} \times \frac{T_b}{2} \left[1 - \frac{2\tau}{T_b} \right] = R_g(\tau)$$

$$R_g(\tau) = \frac{1}{2} \left[1 - \frac{2\tau}{T_b} \right], \quad t \leq \frac{T_b}{2}$$

$$R_g(\tau) = \frac{1}{2} \left[1 - \frac{2|\tau|}{T_b} \right], \quad |\tau| \leq \frac{T_b}{2}$$

even function

(ii) For shift $\tau > \frac{T_b}{2}$



$$R_g(\tau) = 0, \quad \tau > \frac{T_b}{2}$$

↳ because there's +ve and -ve overlapping (50%)

$$\int g(t) g(t-\tau) dt = 0$$

(39)

For book ka
question:

$$\tau < \frac{T_b}{2}$$

$$T = NT_b$$

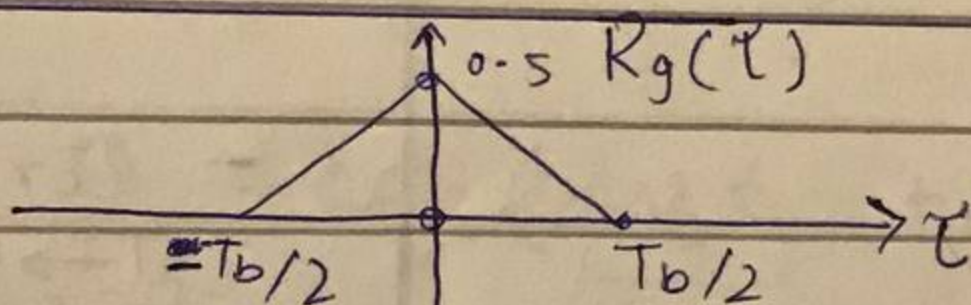
$$\left(\frac{N}{2} \right) \left(\frac{T_b}{2} - \tau \right)$$

because no pulse
for 'zeros' and
no area for '0s'

$$\tau > \frac{T_b}{2}$$

$\neq 0$ because
overlapping with
+1 or '0', so
no cancellation

Example 3.23 Cont.



$$\Delta \left(\frac{t}{T} \right) \longleftrightarrow \frac{T}{2} \text{sinc}^2 \left(\frac{\omega T}{4} \right)$$

$$R_g(t) \longleftrightarrow S_g(\omega)$$

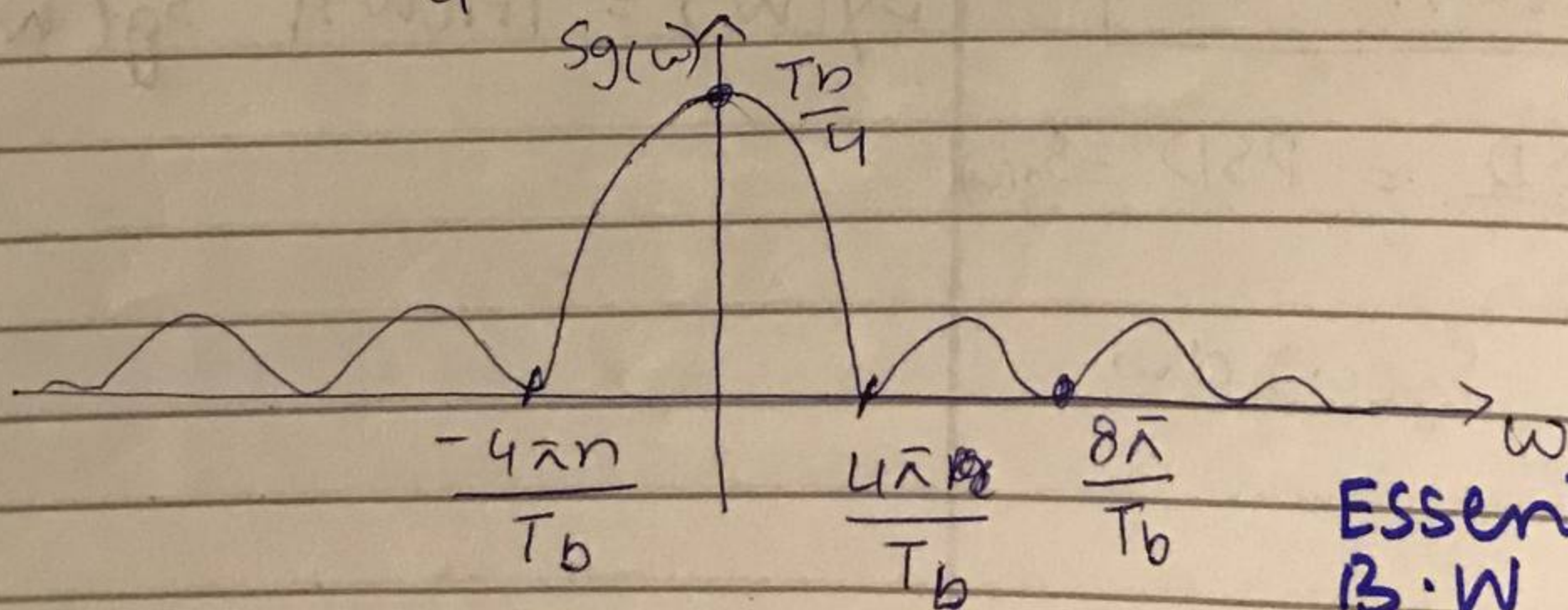
In our case:

$$\Delta \left(\frac{t}{T_b} \right) \longleftrightarrow \frac{T_b}{2} \text{sinc}^2 \left(\frac{\omega T_b}{4} \right)$$

$$S_g(\omega) = \frac{T_b}{4} \text{sinc}^2 \left(\frac{\omega T_b}{4} \right)$$

$$\frac{\omega T_b}{4} = \pm n\pi$$

$$\omega = \pm \frac{4n\pi}{T_b}$$



Essential
B.W is in
first lobe so
E.BW = $4\pi/T_b$ rad/s

Energy signals

(40)

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

$$|G(\omega)|^2 = \text{ESD} = \Psi_g$$

Auto-correlation func. :

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t \pm \tau) dt$$

$$\Psi_g(\tau) \xleftrightarrow{F.T} \text{ESD } \Psi_g(\omega)$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi_g(\omega) d\omega$$

$$\textcircled{1} X(\omega) \rightarrow |X(\omega)|^2 = \text{ESD}$$

$$\textcircled{2} \lim_{T \rightarrow \infty} \frac{\text{ESD}}{T} = \text{PSD} = S_g(\omega)$$

$$\textcircled{3} P = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

Power signals

$$P_g = \lim_{T \rightarrow \infty} \frac{E_g}{T} = \underbrace{g^2(t)}_{\text{mean sq. value}}$$

$$[g(t)]_{\text{rms}} = \sqrt{P_g}$$

$$\text{PSD} = S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G(\omega)|^2}{T}$$

Auto-correlation func.

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} g(t)g(t \pm \tau) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \Psi_g(\tau)$$

$$R_g(\tau) \xleftrightarrow{F.T} S_g(\omega) \text{ PSD}$$

$$\textcircled{P_g} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

$$S_y(\omega) = |H(\omega)|^2 S_g(\omega)$$

Sept 5, 2017

Amplitude (Linear) Modulation

Target: To transfer our signal at long-distances we need to shift base-band signals to Pass-band signals.
 (freq. near 0 Hz) shifting Higher frequency

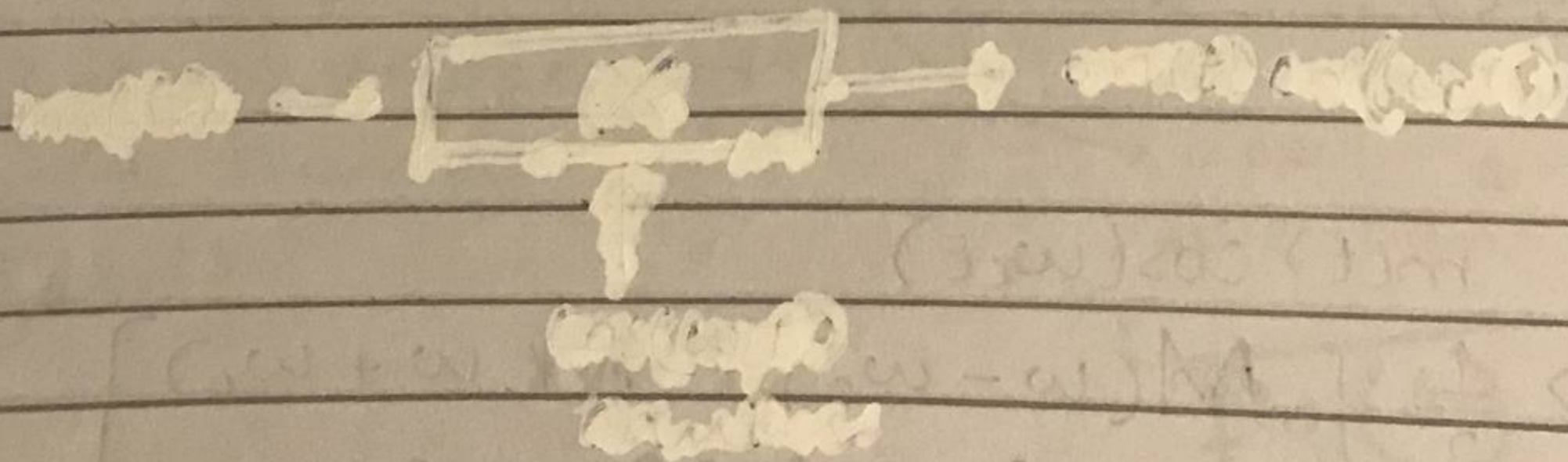
Shape must not be changed, so we change its frequency to a higher one. More than one signal can be simultaneously sent.

$$C(t) = \underbrace{A}_{\text{amp.}} \cos \left(\underbrace{\omega_c t}_{\text{freq. mod.}} + \underbrace{\theta}_{\text{Phase mod.}} \right)$$

Modulation Types:

- Amplitude
 - Phase
 - frequency
- } angle mod.

$$C(t) = A \cos(\theta(t))$$

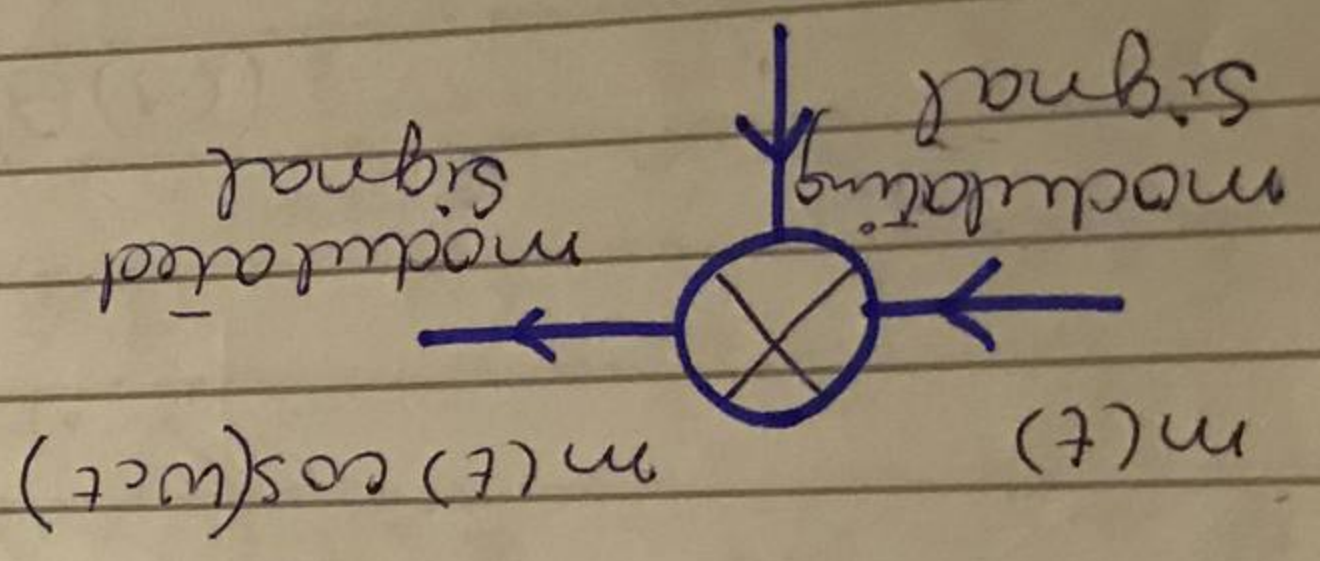


One of the basic parameters (amp, phase) of a sinusoidal carrier of high frequency ω_c is varied in proportion to the baseband signal.

Types of amplitude modulation:

- Double side band suppressed carrier (DSB-SC)
- Double side band with carrier (DSB-FC, AM)
- Single side band
 - SSB + C
 - SSB-SC
- Vestigial side band
 - VSB + C
 - VSB-SC

DSB-SC



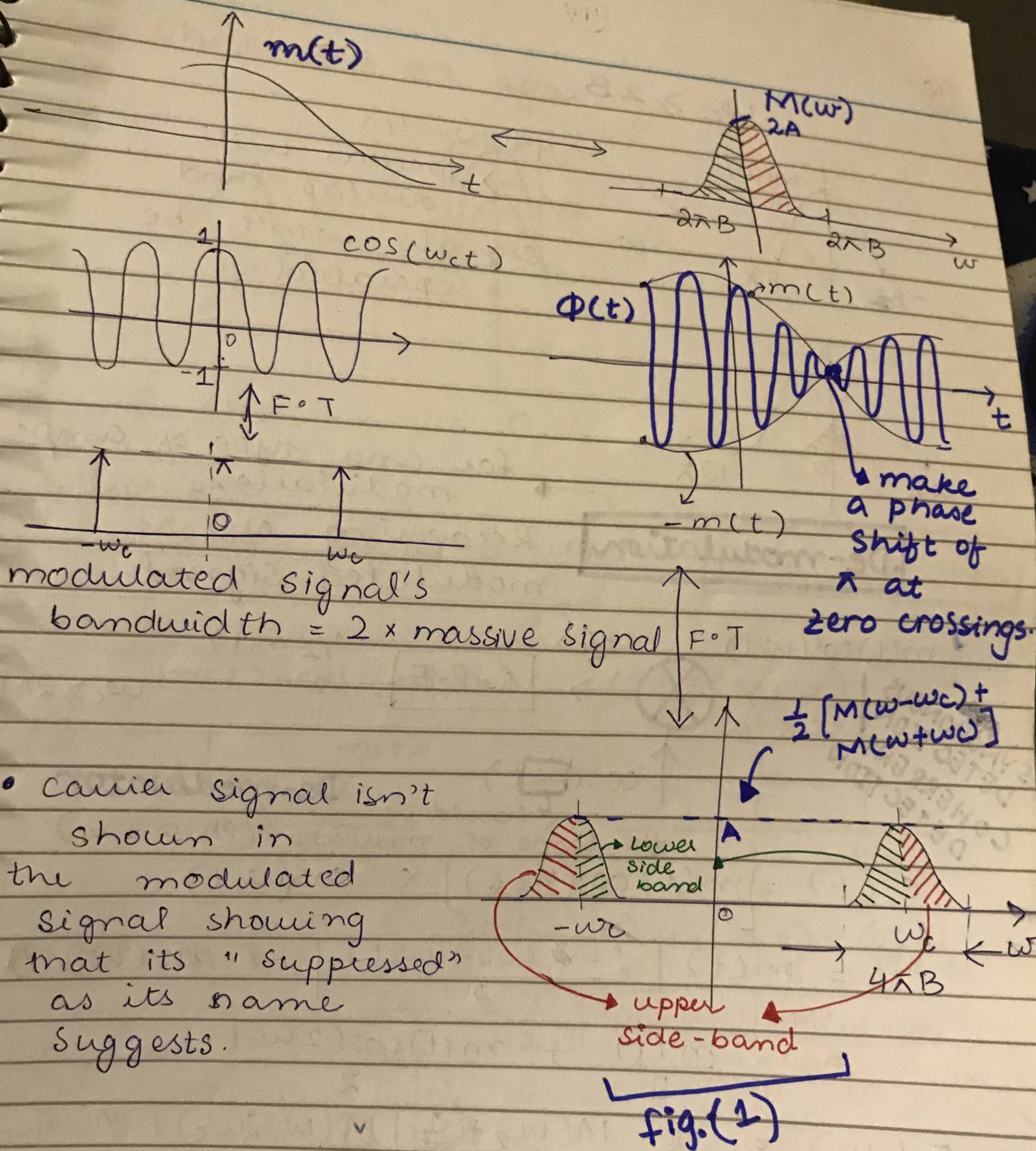
modulating signal / massive signal /
 its signal may change lama, modulate
 karna hai.

$$\phi(t) = m(t) \cos(\omega_c t) \Rightarrow \frac{1}{2} [M(\omega - \omega_c) + M(\omega + \omega_c)]$$

frequency shifting

3

43

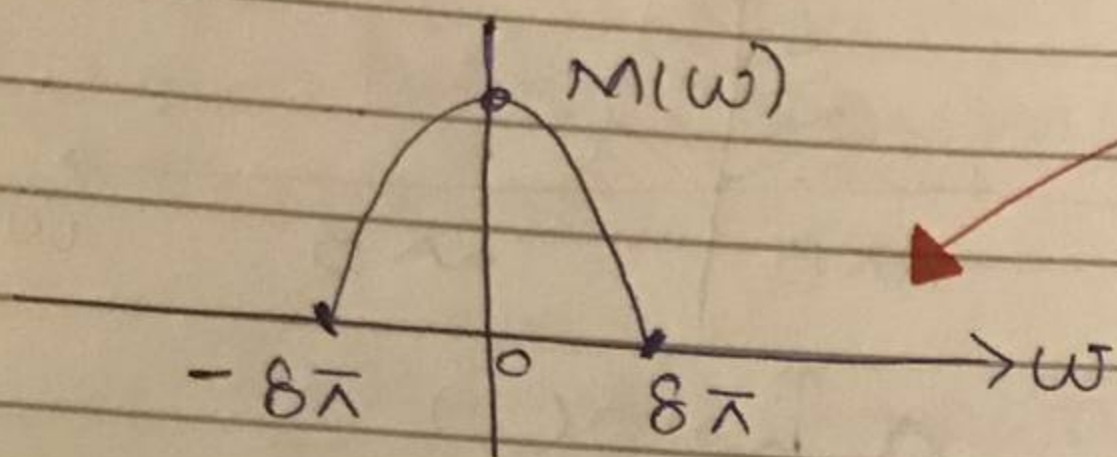


(4)

(44)

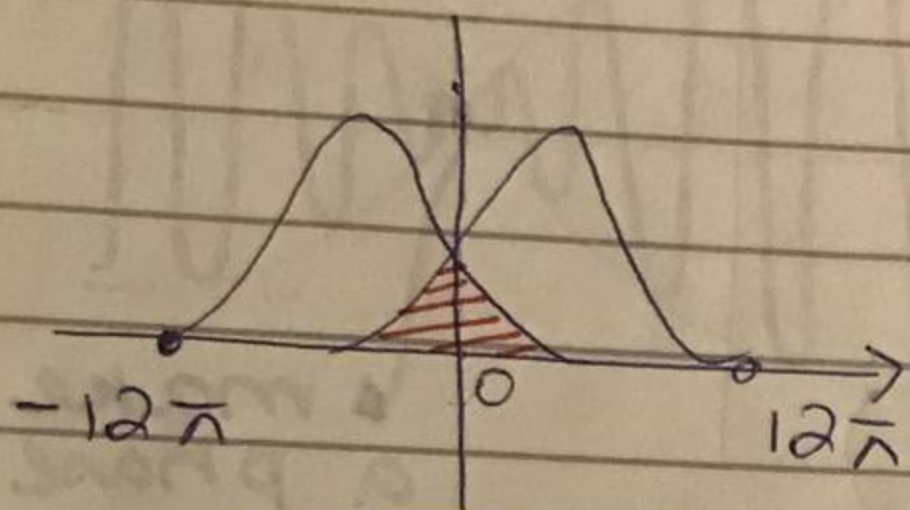
→ causal nhe
 banay ga
 → unrealizable

$$\omega_c \geq 2B$$



if $\omega = 4\pi$

→ spectra will overlap and signals won't be separated



De-modulation

for any type of amp. modulation
 Recovering of modulated signal.

SYNCHRONOUS
 DETECTION /
 COHERENT
 DETECTION

$$m(t) \cos(\omega_c t)$$



$$e(t)$$

$$L \cdot P \cdot F$$

$$\frac{1}{2} m(t)$$

$$\omega_c \geq 2\pi B$$

$$\cos(\omega_c t)$$

De-modulator

should be the same of modulator (Phase + ω)

$$e(t) = [m(t) \cos(\omega_c t)] \times \cos(\omega_c t)$$

$$= m(t) \left[\frac{1 + \cos(2\omega_c t)}{2} \right]$$

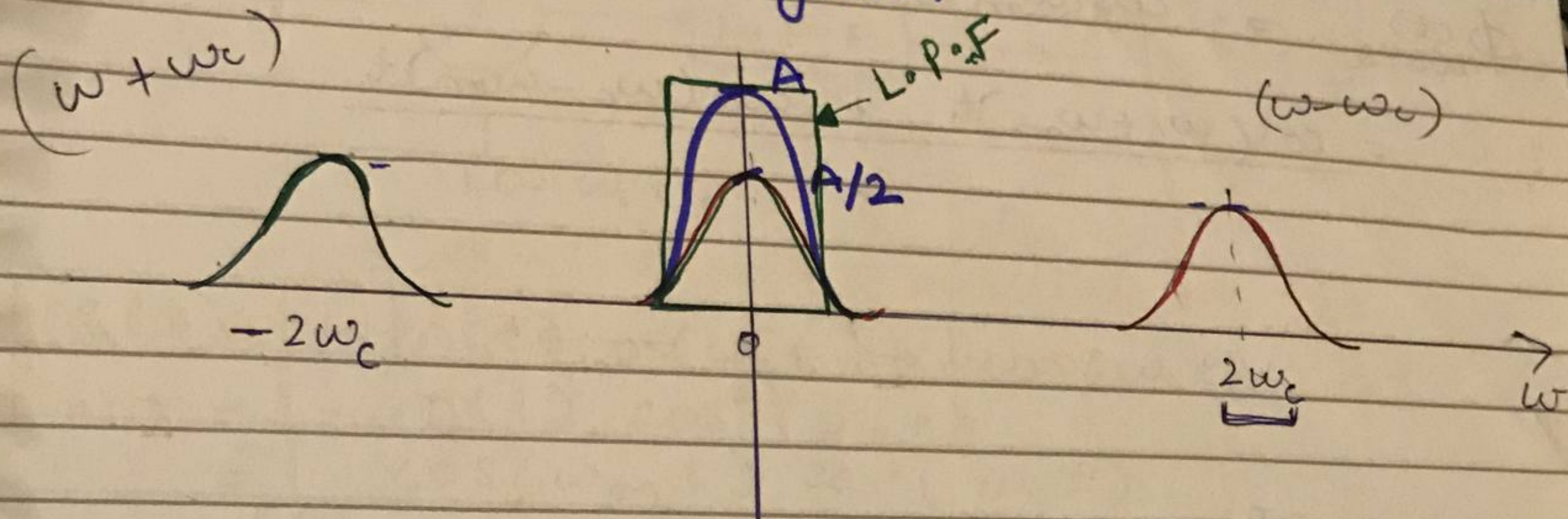
$$= \frac{m(t)}{2} + \frac{m(t) \cos(2\omega_c t)}{2}$$

$$E(\omega) = \frac{1}{2} M(\omega) + \frac{1}{4} [M(\omega - 2\omega_c) + M(\omega + 2\omega_c)]$$

5

45

Shift fig. 4) again



2 Spectrums at 0 (add)

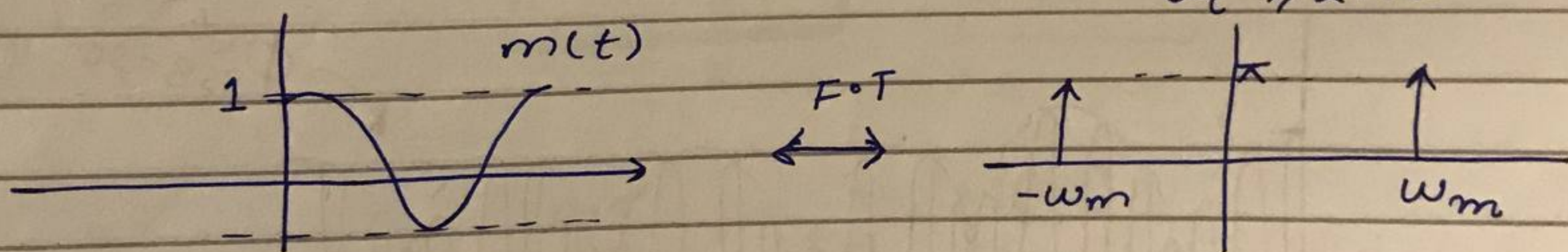
7-Sept-2017

Massive signal = sinusoid $\rightarrow$ Tone modulation

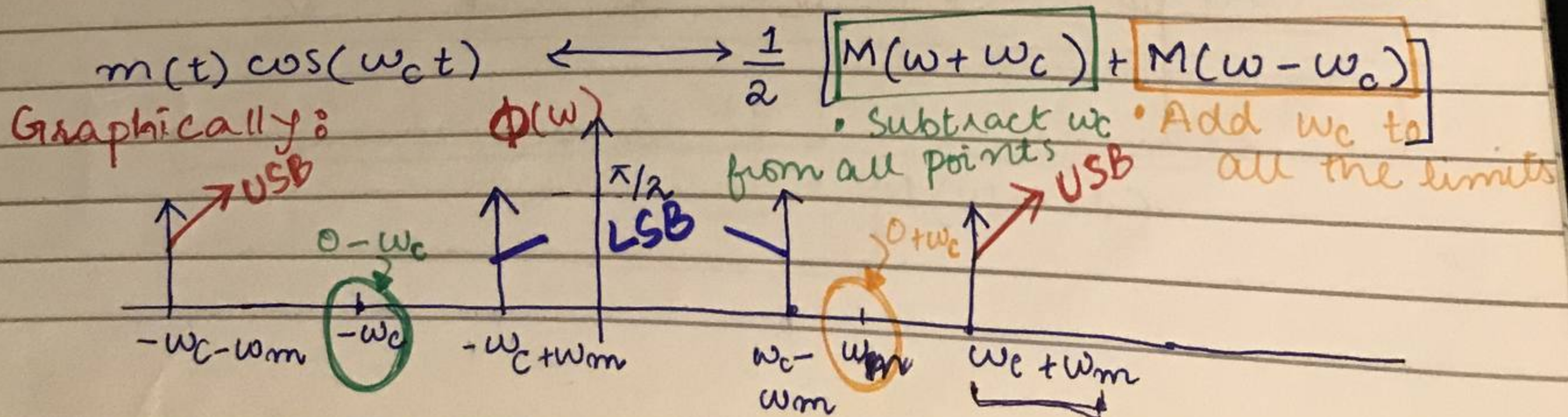
Example 4.1 :

$$m(t) = \cos(\omega_m t), \quad c(t) = \cos(\omega_c t)$$

$$\omega_c \gg 2\pi B$$



$$m(t) \cos(\omega_c t) = \cos(\omega_m t) \cos(\omega_c t) = \phi(t) \quad \text{DSB-SC}$$



(b)

(46)

Mathematically:

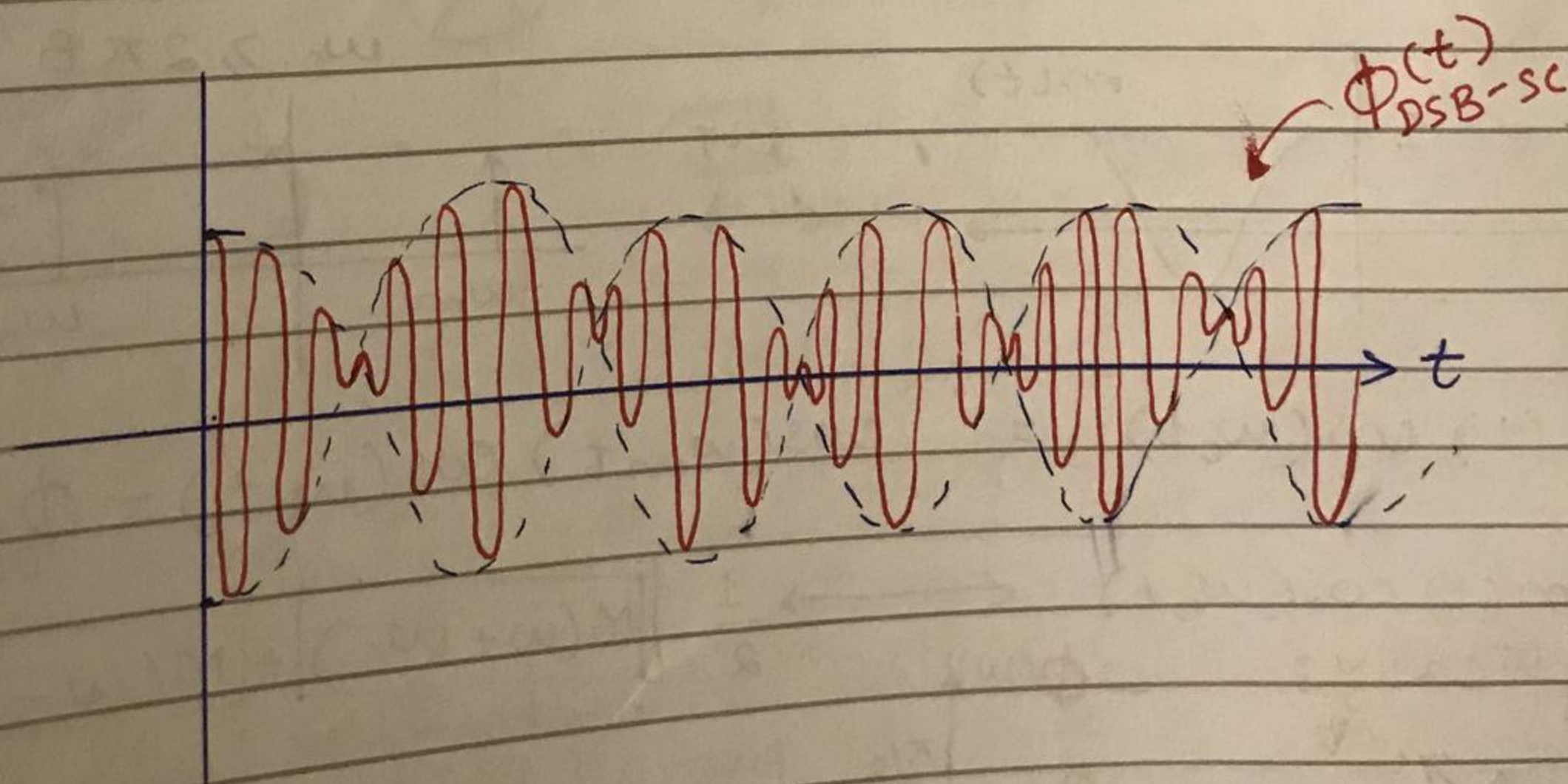
$$\phi_{\text{DSB-SC}}(t) = \cos(\omega_m t) \cos(\omega_c t)$$

$$\phi_{\text{DSB-SC}}(t) = \frac{\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t}{2}$$

$$\cos(\omega_c + \omega_m)t \longleftrightarrow \pi \left\{ \delta[\omega + (\omega_c + \omega_m)] + \delta[\omega - (\omega_c + \omega_m)] \right\}$$

$$\phi_{\text{DSB-SC}}(t) \xleftrightarrow{F.T} \frac{\pi}{2} \left[\delta(\omega + \omega_c + \omega_m) + \delta(\omega - (\omega_c + \omega_m)) \right] + \frac{\pi}{2} \left[\delta(\omega + (\omega_c - \omega_m)) + \delta(\omega - (\omega_c - \omega_m)) \right] = \phi(\omega)$$

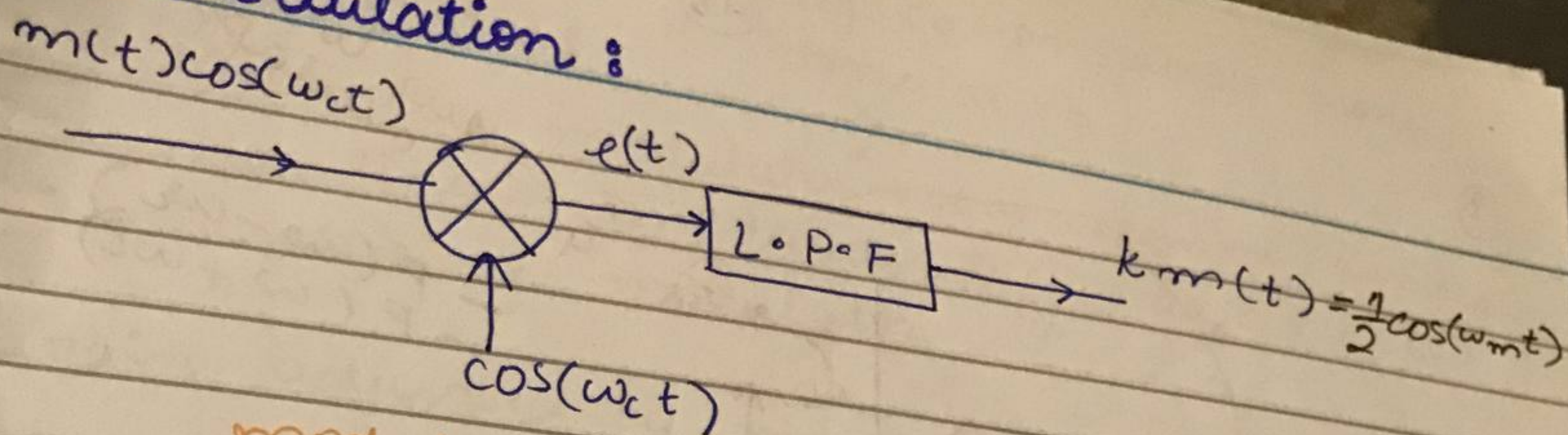
$\Rightarrow$ Graphically = mathematically



(7)

(47)

Demodulation:



modulated signal $\times$ carrier signal at receiver end

$$\begin{aligned}
 e(t) &= m(t) \cos(\omega_c t) \times \cos(\omega_c t) \\
 &= m(t) \cos^2(\omega_c t) \\
 &= \cos(\omega_m t) \cos^2(\omega_c t) \\
 &= \cos(\omega_m t) \left[\frac{1 + \cos(2\omega_c t)}{2} \right]
 \end{aligned}$$

$$= \frac{\cos(\omega_m t)}{2} + \frac{\cos(\omega_m t) \cos(2\omega_c t)}{2}$$

Add ① & ②

use this as reference

$$m(t) \cos(\omega_m t) \times \cos(2\omega_c t)$$

① $\cos(\omega_m t) \xleftrightarrow{F \cdot T} \pi [\delta(\omega - \omega_m) + \delta(\omega + \omega_m)]$

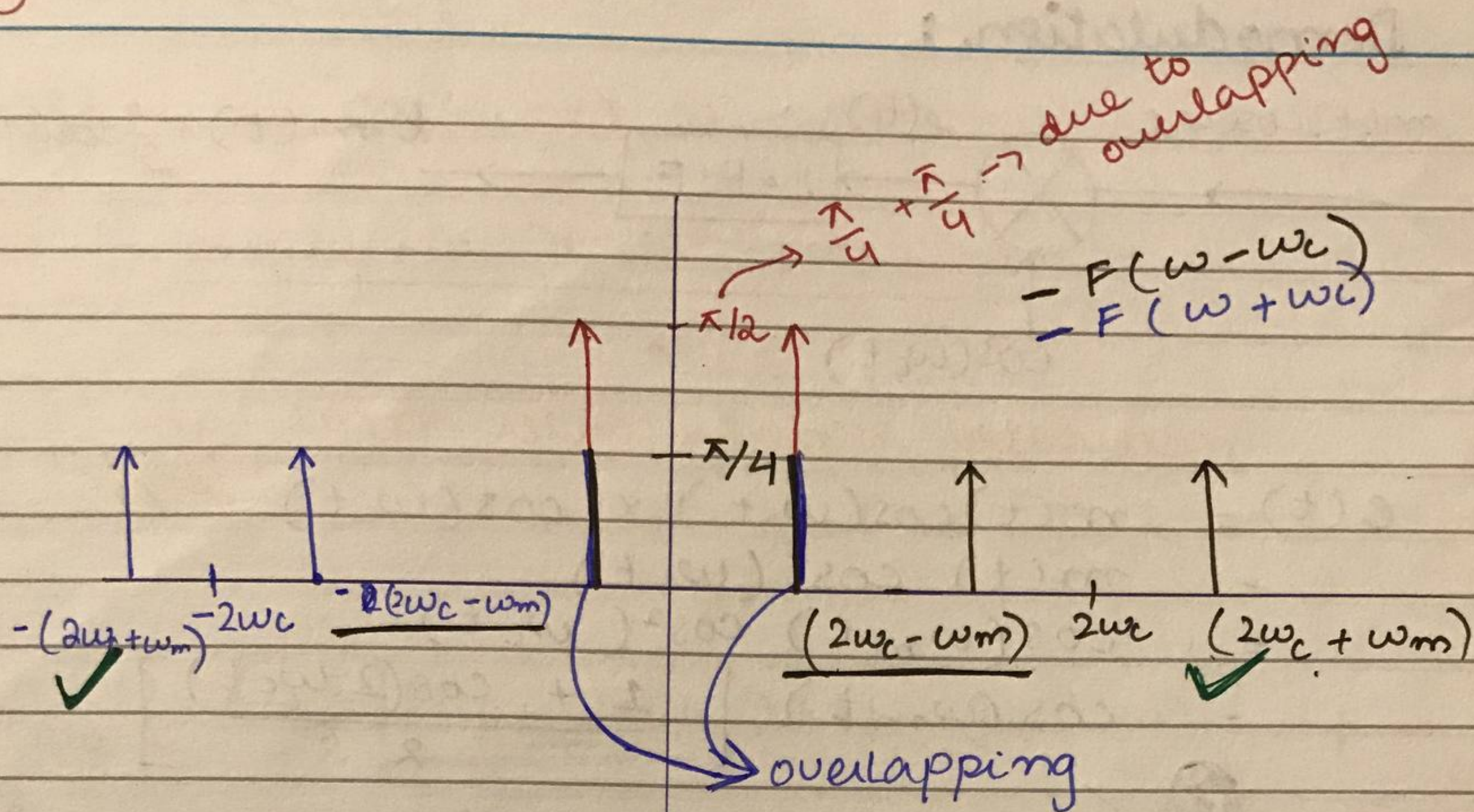
② $\cos(\omega_m t) \cdot \cos(2\omega_c t) \xleftrightarrow{F \cdot T} \frac{1}{2} \pi [\delta(\omega - \omega_m + 2\omega_c) + \delta(\omega + \omega_m + 2\omega_c) + \delta(\omega - \omega_m - 2\omega_c) + \delta(\omega + \omega_m - 2\omega_c)]$

$M(\omega + \omega_1)$

$M(\omega - \omega_1)$

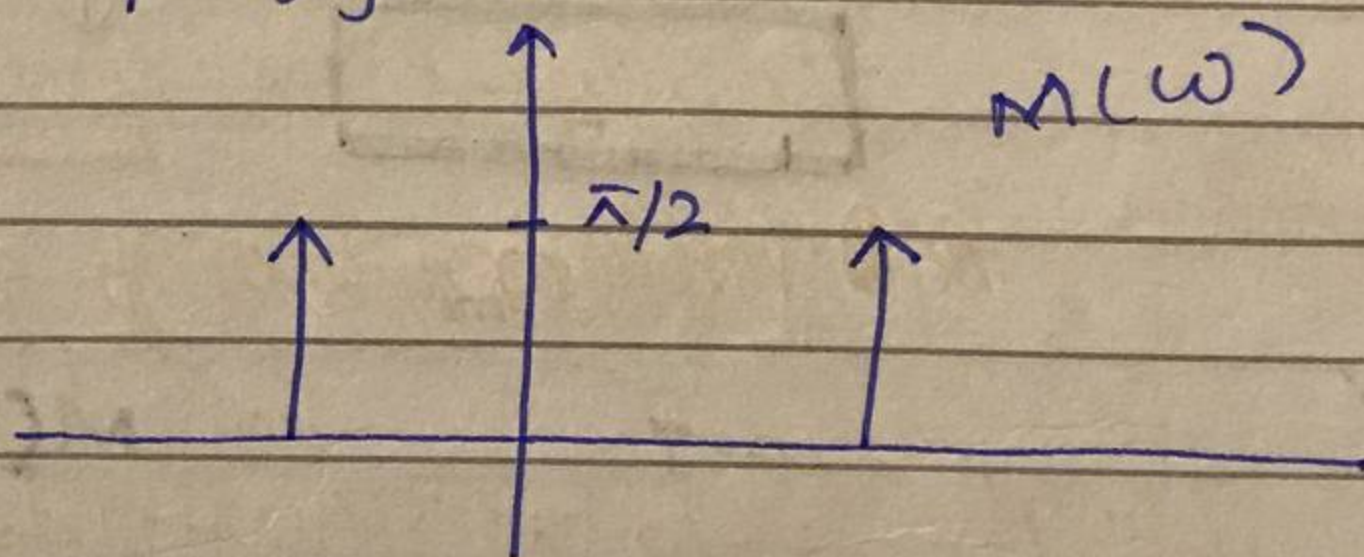
(8)

(48)



Pass it through L.P.F :

$\Rightarrow$ Amplitude mein attenuation hai
 after L.P.F. to koi baat nhe,
 we can amplify it later.



(49) For the multiplication process done in modulation.

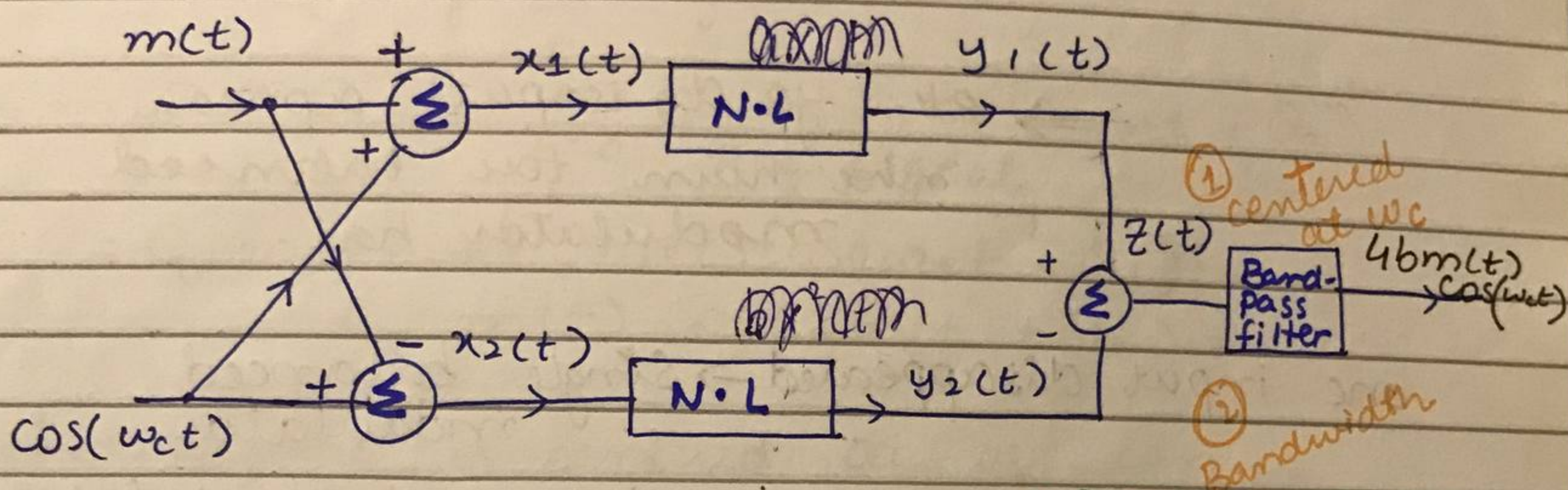
⑨

Modulators:

- 1 - Multiplier modulator $\rightarrow$ very expensive (avoided)
- 2 - Non-linear modulator $\rightarrow y(t) = ax(t) + bx^2(t)$
- 3 - Switching modulator

Gain adjusted by massive signal.

$$x(t) \rightarrow [N \cdot L] \rightarrow y(t)$$



$\rightarrow$ Single modulator w.r.t carrier signals

Maclaurin series represents non-linearity

$$x_1(t) = \cos(\omega_c t) + m(t)$$

$$x_2(t) = \cos(\omega_c t) - m(t)$$

$$ax_1(t) + bx_2^2(t)$$

$$y_1(t) = a[\cos(\omega_c t) + m(t)] + b[\cos(\omega_c t) + m(t)]^2$$

$$y_2(t) = a[\cos(\omega_c t) - m(t)] + b[\cos(\omega_c t) - m(t)]^2$$

$$y_1(t) = a \cos(\omega_c t) + a m(t) + b \cos^2(\omega_c t) + b m^2(t) + 2bm \cos(\omega_c t)$$

$$y_2(t) = a \cos(\omega_c t) - a m(t) + b \cos^2(\omega_c t) + b m^2(t) - 2bm \cos(\omega_c t)$$

$$y_1(t) - y_2(t) = z(t)$$

$$z(t) = 2am(t) + 4bm(t)\cos(\omega_c t)$$

need to pass through B.P filter

B.P filter centered at ω_c

(10)

* If relationship of linear device is

$$y(t) = bx^2(t)$$

then

$$z(t) = 4bm(t) \cos(\omega_c t)$$

↳ balanced w.r.t
input and output

⇒ ek ya do inputs appear
mhe hum to balanced
modulator hai.

one input disappeared → Single balanced
modulator

both inputs gone → Double balanced modulator

12 - Sept - 2017

4.2

Switching Modulators:

$$\phi(t) = \sum_{n=0}^{\infty} C_n \cos(n\omega_c t + \theta_n) \quad \text{--- (i)}$$

$$\phi_{DSB-sc}^{(t)} = m(t) \cos(\omega_c t)$$

multiplication ka operation can be replaced with a switch.

multiply eq. (i) by $m(t)$

$$m(t) \phi(t) = \sum_{n=0}^{\infty} C_n m(t) \cos(n\omega_c t + \theta_n)$$

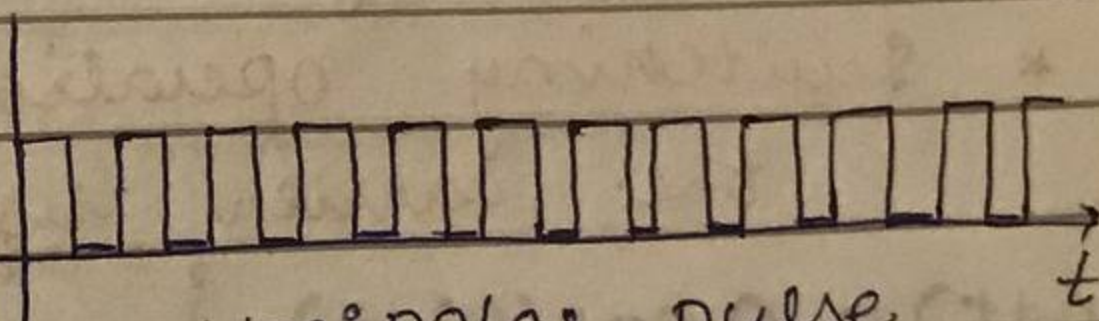
Req. signal

$$= C_0 m(t) + C_1 m(t) \cos(\omega_c t + \theta_1) + C_2 m(t) \cos(2\omega_c t + \theta_2) + \dots$$

Req. signal is centered at ω_c can be extracted using bandpass filter

→ BPF centered at $\pm \omega_c$ with $2B$ Hz Bandwidth.

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \left[\cos(\omega_c t) - \frac{1}{3} \cos(3\omega_c t) + \frac{1}{5} \cos(5\omega_c t) \dots \right]$$

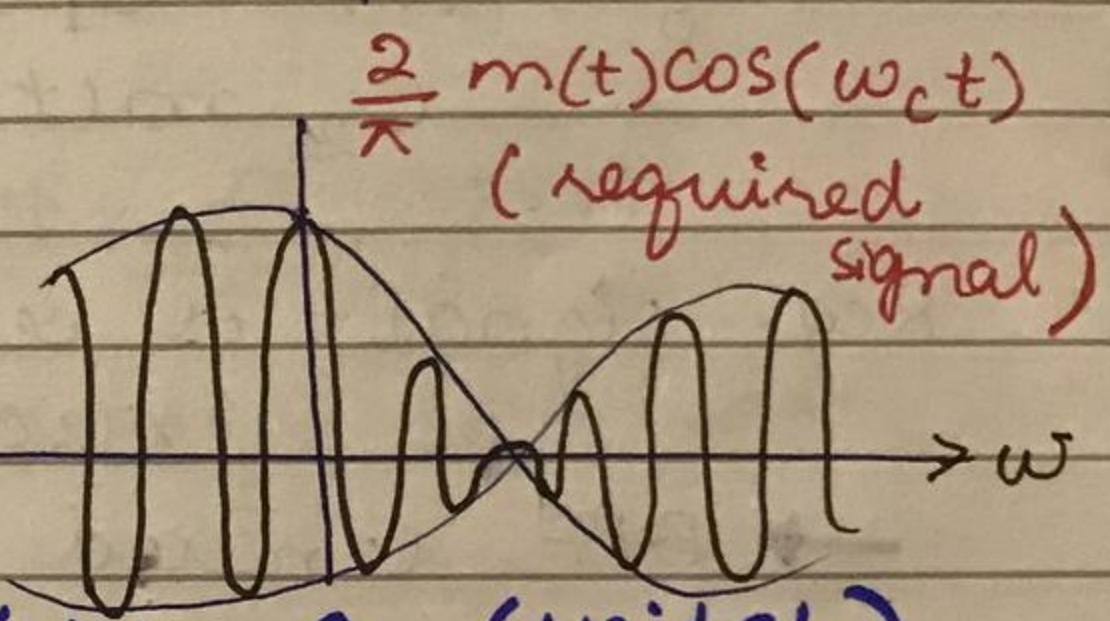
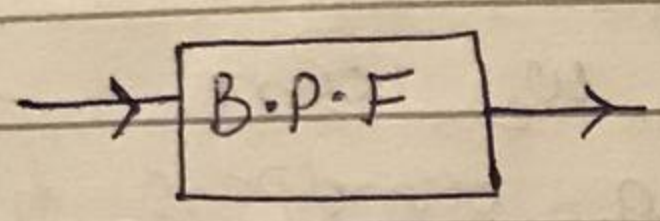
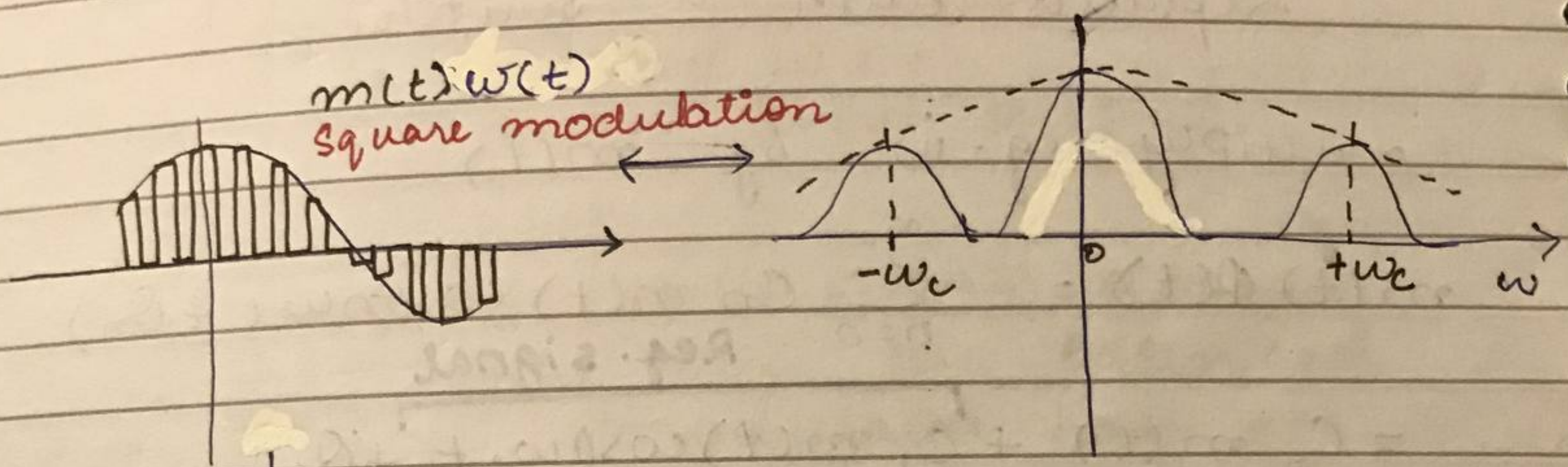
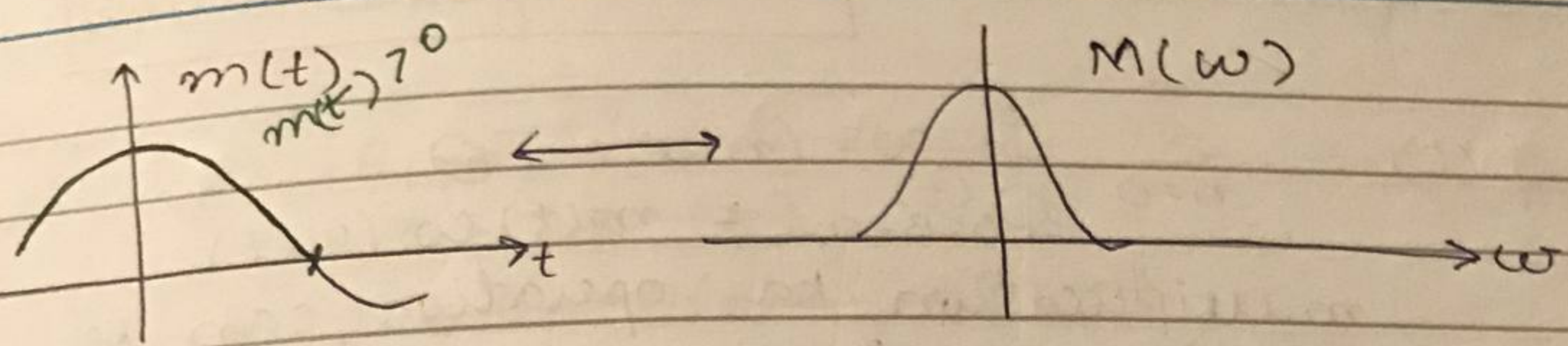


unipolar pulse train $w(t)$
we want to extract this term

$$m(t) w(t) = \frac{m(t)}{2} + \frac{2}{\pi} m(t) \cos(\omega_c t) - \frac{2}{3\pi} m(t) \cos(3\omega_c t) + \frac{2}{5\pi} m(t) \cos(5\omega_c t) + \dots$$

$$m(t) w(t) \rightarrow \boxed{\text{BPF at } \pm \omega_c} = \frac{2}{\pi} m(t) \cos(\omega_c t)$$

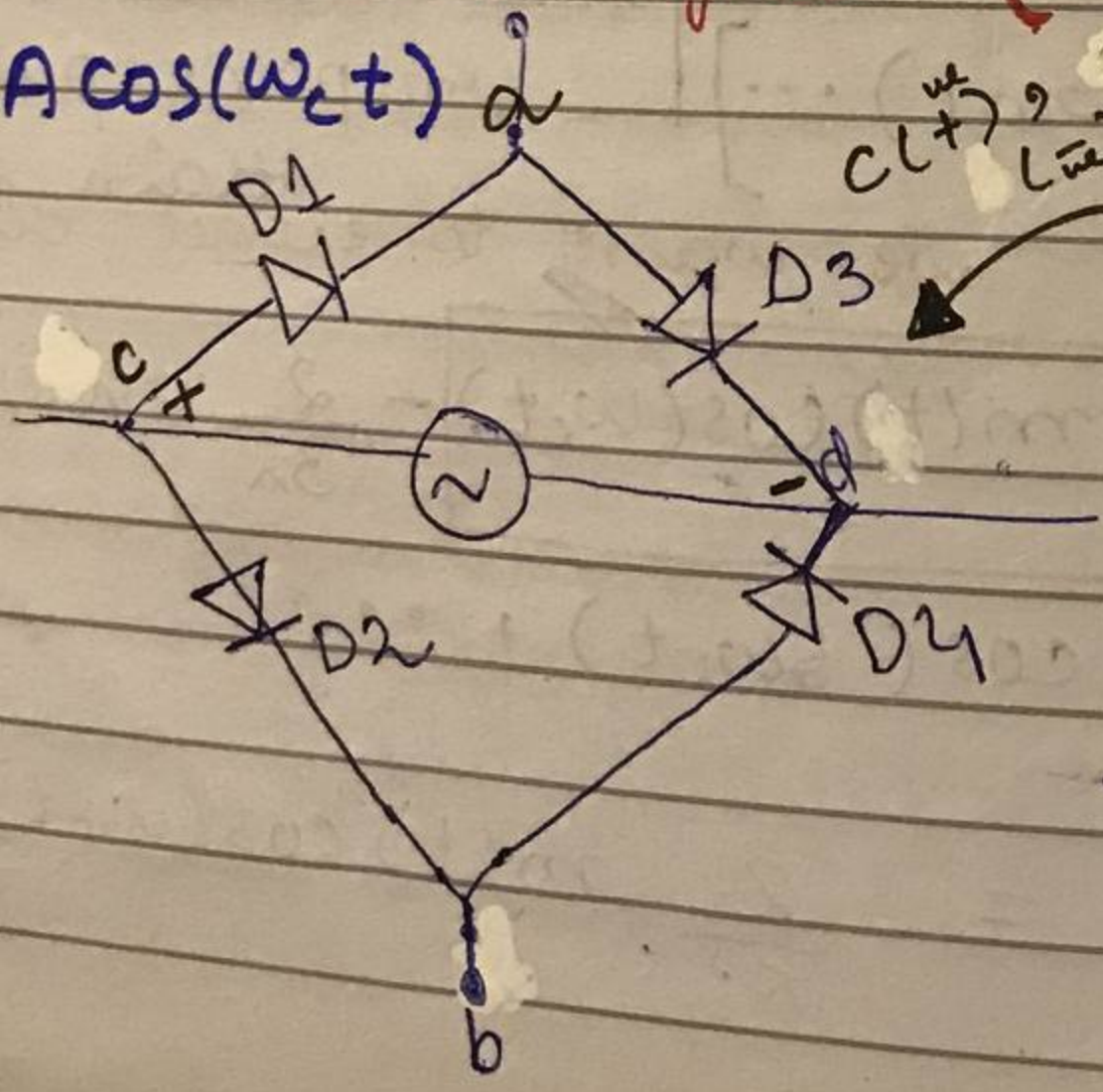
(12)



Diode Bridge Modulator (as a switch)

* Switching operation is controlled by the carrier signal (when to off & on)

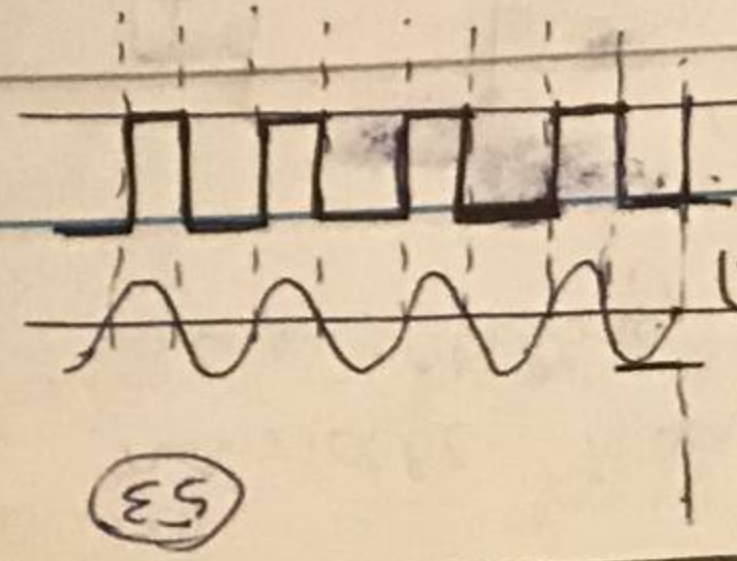
$$c(t) = A\cos(\omega_c t)$$



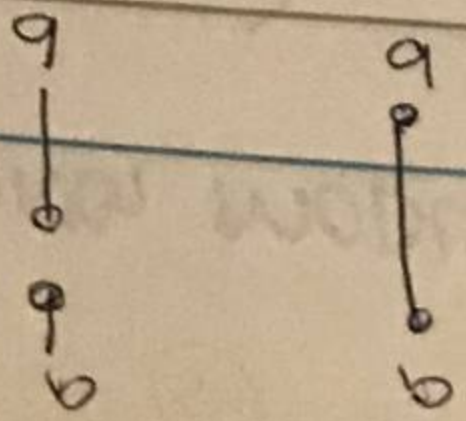
+ve cycle (closed circuit)
all diodes are conducting (ON)

-ve cycle (open circuit)

Diode Bridge Modulator
or a
Switch Electronic



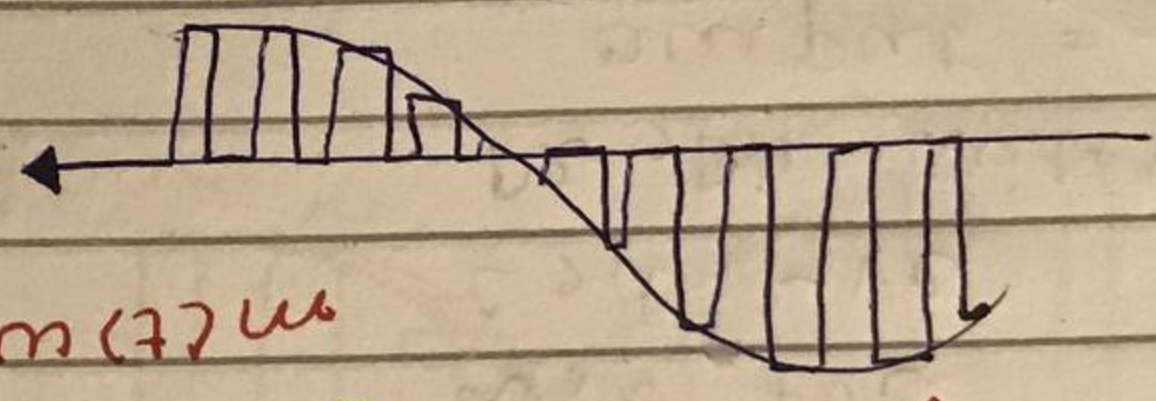
(13)



Shunt = Parallel

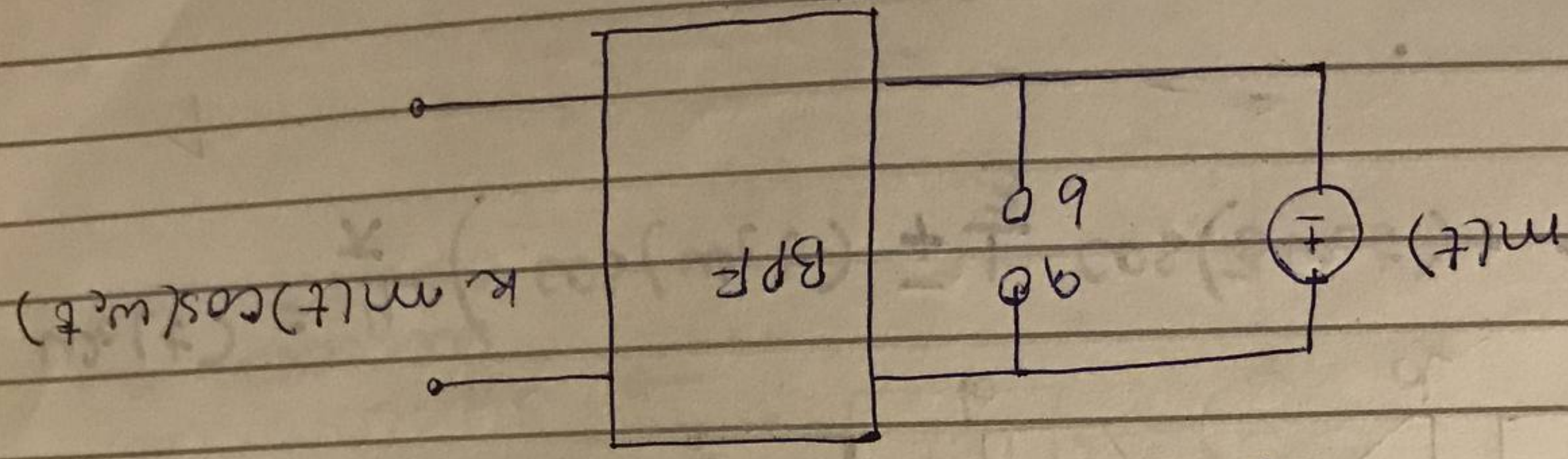
Series Bridge Diode Modulator
(+ve cycle — signal logic 1
-ve cycle — logic 0)

$m(t)w(t)$



Parallel Bridge Diode Modulator:

-ve cycle — logic 1
+ve cycle — logic 0

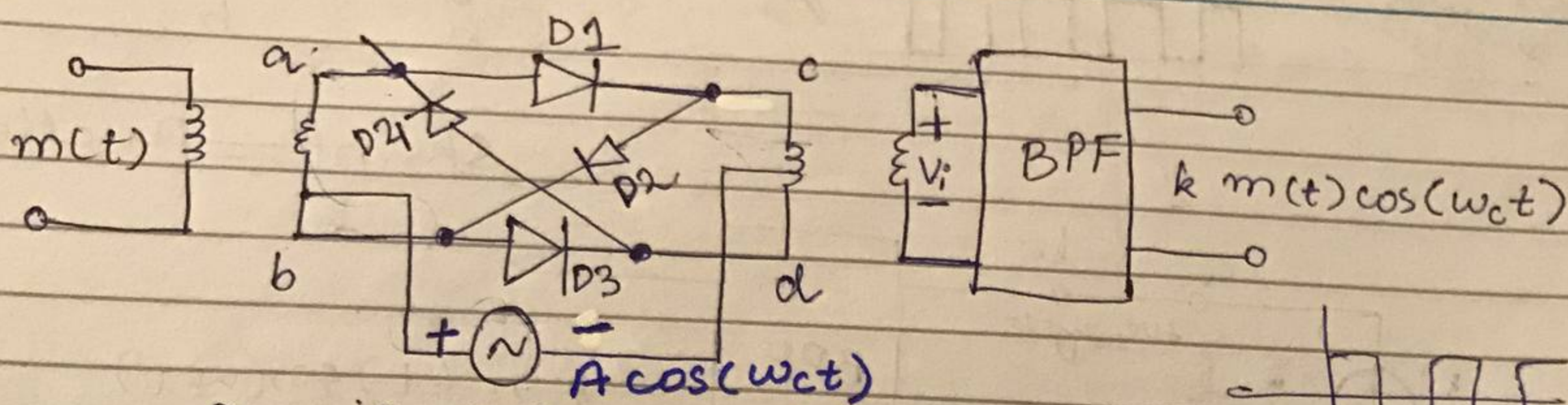


(14)

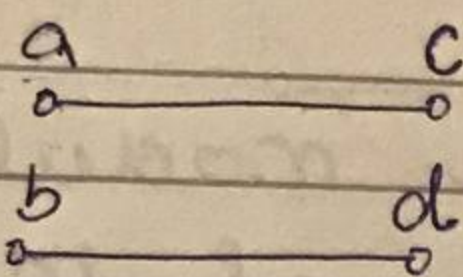
(54)

RING MODULATOR:

carrier handle kar raha hai
switching operator



a is +ve w.r.t to c



$C(t) > 0$ +ve cycle
output = $m(t)$

D1, D3 (on)

D2, D4 (off)

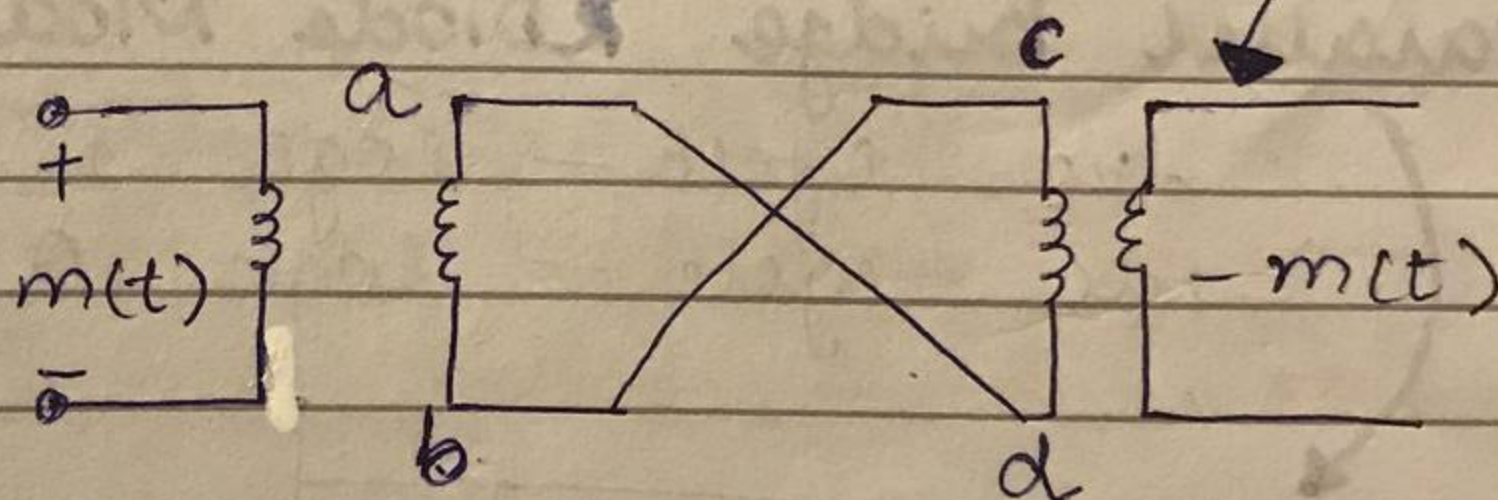
$C(t) < 0$
in -ve cycle

a, b +ve

c, d -ve

D2, D4 will conduct

output = $-m(t)$



$$w_o(t) = \frac{4}{\pi} \left(\cos(\omega_c t) - \frac{1}{3} \cos(3\omega_c t) + \dots \right)$$

common input disappeared → Double balanced modulator

$$m(t) w_o(t) = \frac{4}{\pi} m(t) \cos(\omega_c t) + \frac{4}{\pi} m(t) \cos(3\omega_c t) + \dots$$

neither $m(t)$ nor $\cos(\omega_c t)$

left shifted $\rightarrow$ advanced
 right shifted $\rightarrow$ delayed

(15)

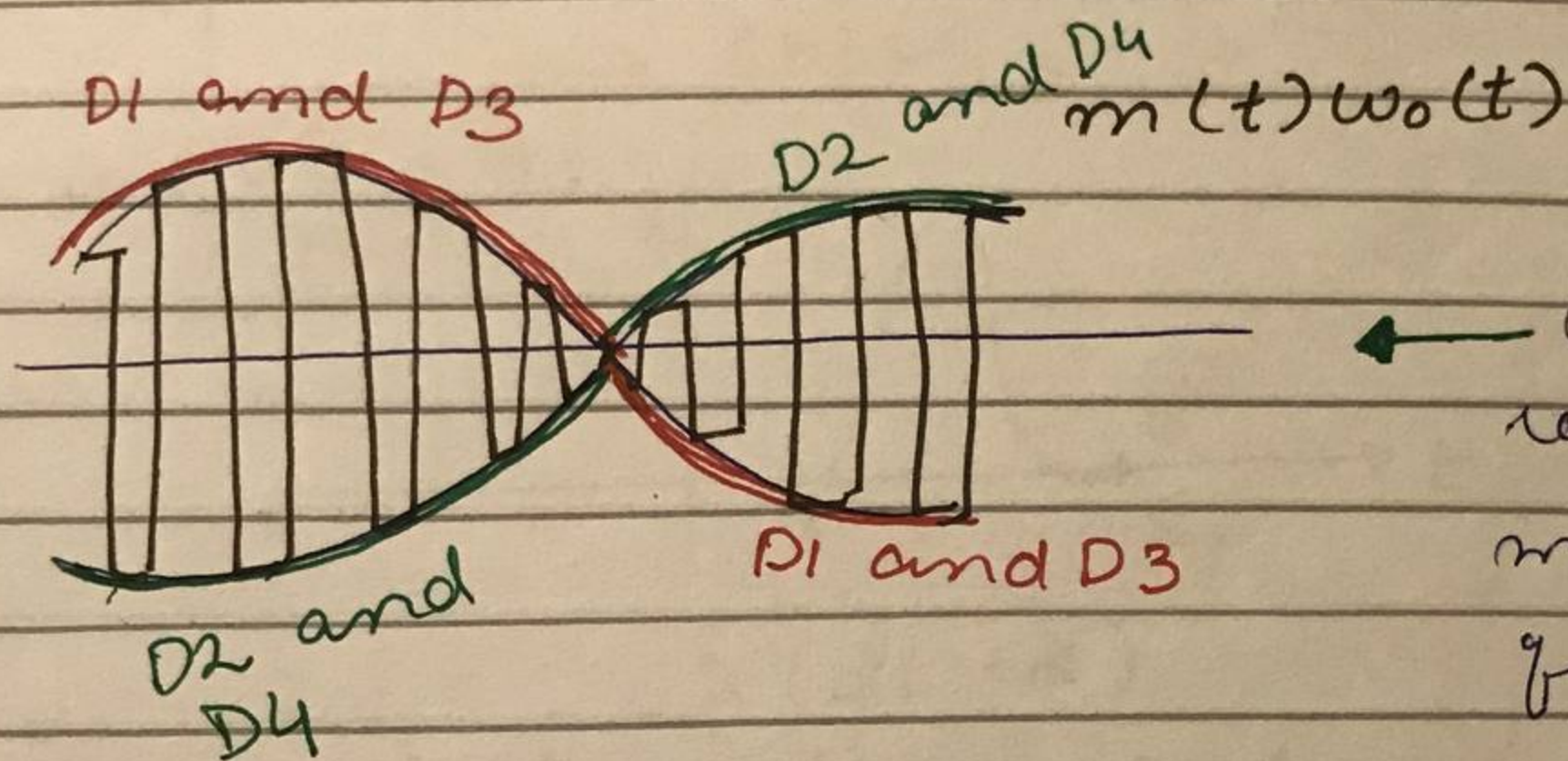
(55)

* $x(2t-3)$ $\rightarrow$ ① shift of 3 in t (right)
 $\rightarrow$ ② scaled t by 2

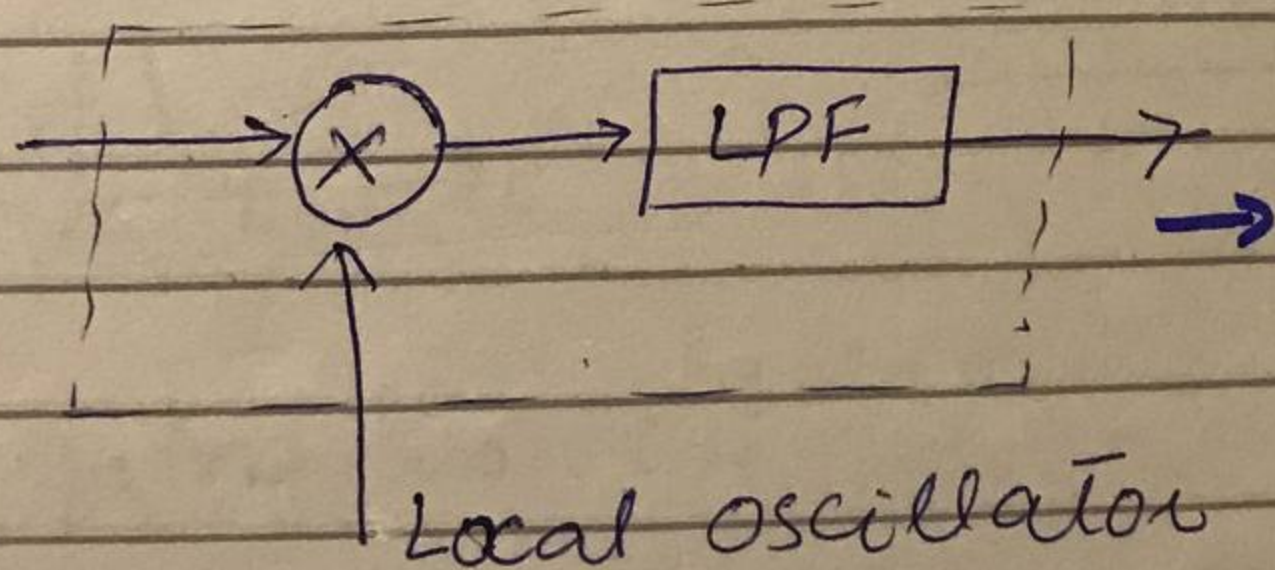
easy to me.

OR

* $x(2(t - \frac{3}{2}))$ ① $\rightarrow$ scaled by 2
 ② $\rightarrow$ shifted by $\frac{3}{2}$ (right)



Graphical
 result of
 multiplication
 for $c(t) > 0$
 and $c(t) < 0$



synchronous
 detector as
 demodulator

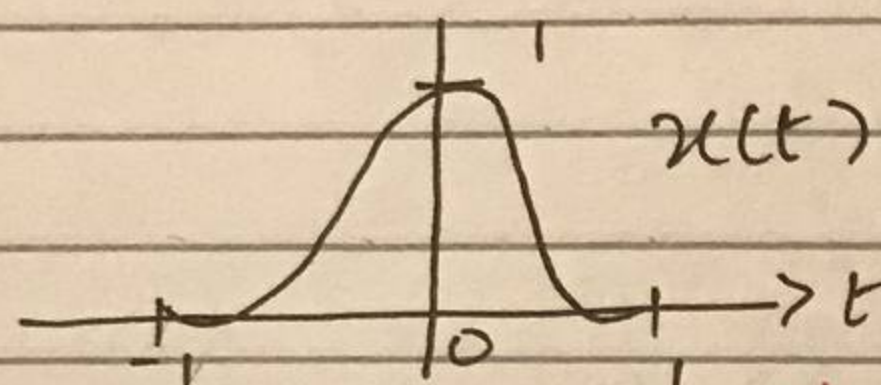
$x(2(t-1))$ $\left\{ \begin{array}{l} \text{① scale, shift} \\ \text{② shift, scale.} \end{array} \right.$

(56)

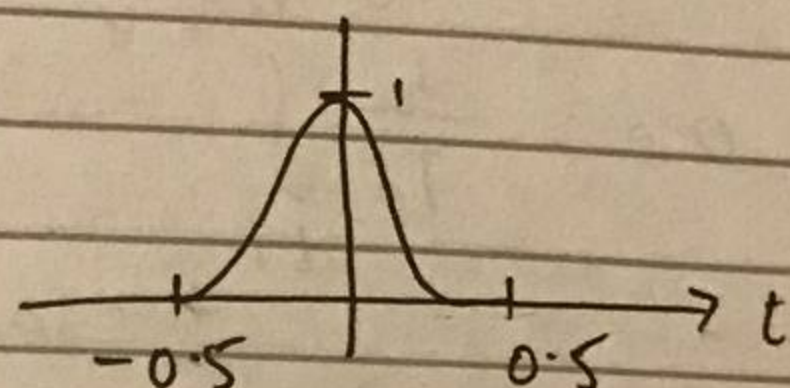
Semester 4 Revision

① $x(t) \rightarrow x(2 \times t)$ **scale first, then shift**
compress the signal by 2
then shift in t $x(2(t-1))$

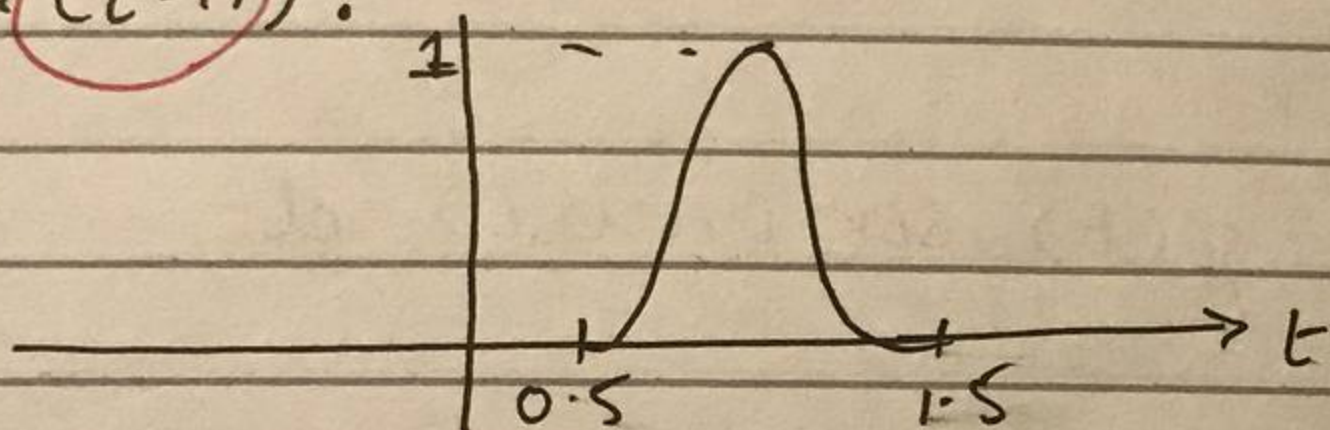
e.g.



$x(2t)$



$x(2(t-1))$: **shift in t**



$$-0.5 + 1 = 0.5 \text{ (L.L.)}$$

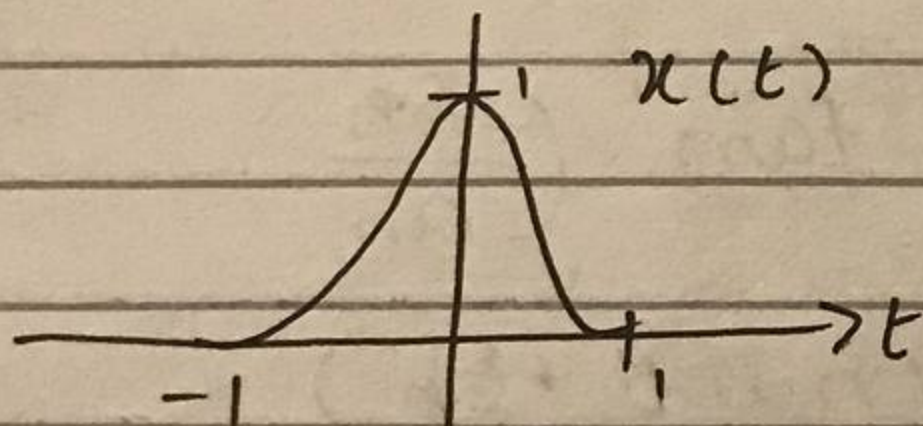
$$0.5 + 1 = 1.5 \text{ (U.L.)}$$

upper limit

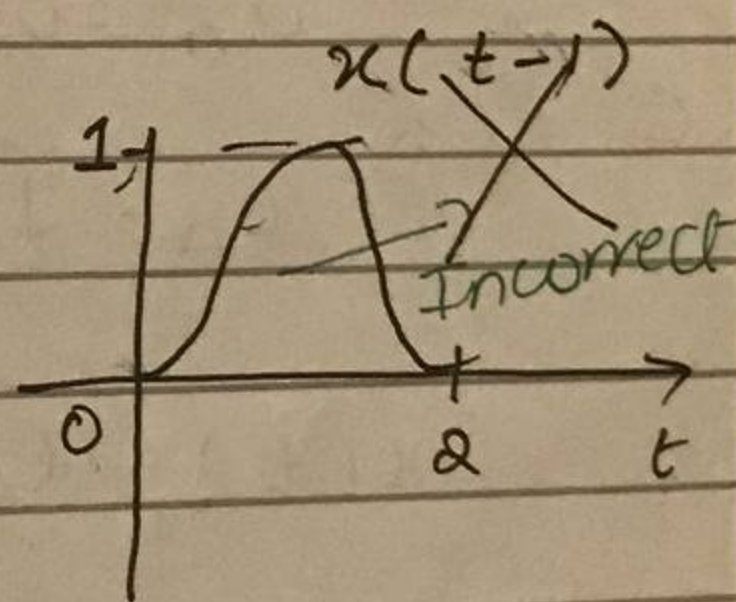
in t

$x(2t-2)$

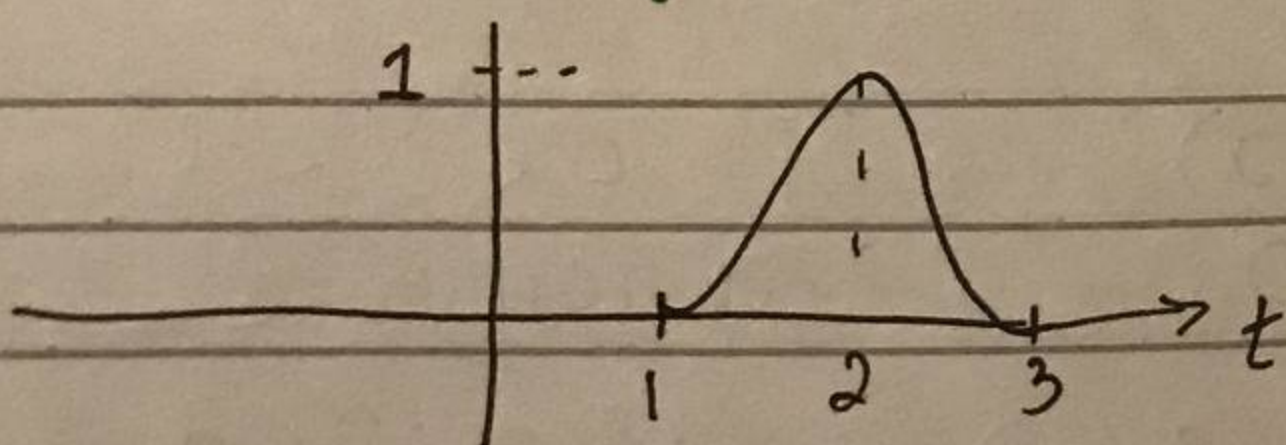
② **Shift first, then scale.**



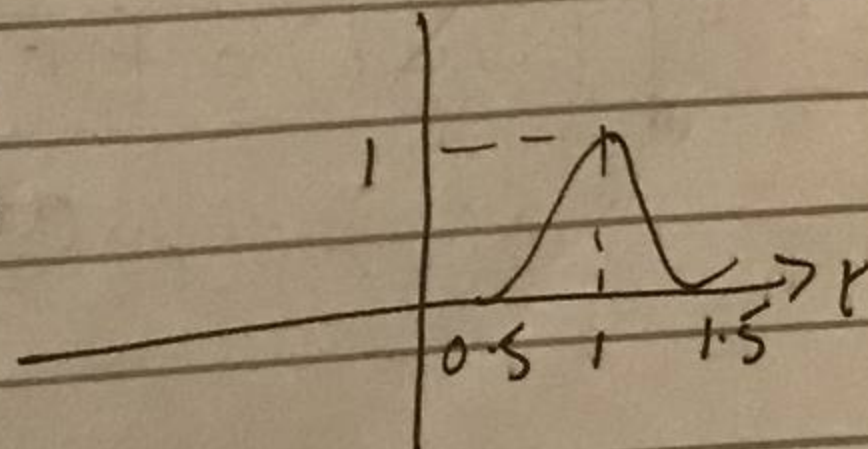
shifting in t



shifting in t by 2



scaling by 2



Trigonometric Fourier Series:

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

$$\omega_0 = \frac{2\pi}{T_0}$$

$$t_1 \leq t \leq t_1 + T_0$$

$$A_0 = \frac{1}{T_0} \int_{t_1}^{t_1 + T_0} x(t) dt$$

$$A_n = \frac{2}{T_0} \int_{t_1}^{t_1 + T_0} g(t) \cos(n\omega_0 t) dt \quad n = 1, 2, 3, \dots$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1 + T_0} g(t) \sin(n\omega_0 t) dt \quad n = 1, 2, 3, \dots$$

compact trigonometric Fourier Series:

$$A_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = C_n \cos(n\omega_0 t + \Theta_n)$$

$$C_0 = A_0$$

$$C_n = \sqrt{A_n^2 + b_n^2}$$

$$\Theta_n = \tan^{-1} \left(\frac{-b_n}{A_n} \right)$$

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \Theta_n)$$

$$t_1 < t \leq t_1 + T_0$$

$$f(t) \delta(t) = f(0) \delta(t)$$

$$\int_c^d \delta(t) f(t) dt = \begin{cases} f(0) & \text{if } c < 0 < d \\ 0 & \text{otherwise} \end{cases}$$

$$\int_c^d \delta(t-a) f(t) dt = \begin{cases} f(a) & \text{if } c < a < d \\ 0 & \text{otherwise} \end{cases}$$

$$f(t) \delta(t-a) = f(a) \delta(t-a)$$